

Dr. Doc:

Exam:-Microbiology

Year:-2014 E.C

Dr. Doc

1.Which statement is not true about *Corynebacterium diphtheriae*?

- A. Causes lower respiratory infection
- B. Causes upper respiratory infection
- C. Secretes an exotoxin that inhibits protein synthesis
- D. Forms a pseudomembrane on the pharynx

Answer A. Causes lower respiratory infection

Corynebacterium diphtheriae is a Gram-positive, club-shaped bacillus that primarily infects the upper respiratory tract, especially the pharynx, tonsils, and larynx. It spreads through respiratory droplets and causes diphtheria. The organism produces a potent exotoxin that inhibits protein synthesis by inactivating elongation factor-2 (EF-2), leading to tissue necrosis. A characteristic feature of the disease is the formation of a thick gray pseudomembrane over the pharynx, which can obstruct the airway if severe. Although the infection may occasionally extend downward, it is not primarily a lower respiratory tract infection.

Why Not Other Options?

B. Causes upper respiratory infection ❌

Correct. *C. diphtheriae* commonly infects the upper respiratory tract, producing sore throat, fever, and cervical lymph node enlargement ("bull neck").

C. Secretes an exotoxin that inhibits protein synthesis ❌

Correct. The diphtheria toxin ADP-ribosylates EF-2, blocking protein synthesis and causing cell death. The toxin can also damage the heart (myocarditis) and peripheral nerves (neuritis).

D. Forms a pseudomembrane on the pharynx ❌

Correct. The hallmark of respiratory diphtheria is the formation of a gray, tightly adherent pseudomembrane over the pharynx or tonsils. Attempting to remove it may cause bleeding.

2. Bacitracin sensitivity is to *S. pyogenes* and Optochin sensitivity is to:

A. *Bordetella pertussis*

B. *Bacillus anthracis*

C. *Streptococcus pneumoniae*

D. *Mycobacterium tuberculosis*

Answer C. *Streptococcus pneumoniae*

Optochin sensitivity is an important laboratory test used to identify *Streptococcus pneumoniae*. *S. pneumoniae* is an alpha-hemolytic, Gram-positive diplococcus that is sensitive to optochin and bile soluble. These characteristics help distinguish it from viridans streptococci, which are also alpha-hemolytic but are optochin-resistant and bile-insoluble. Similarly, *Streptococcus pyogenes* (Group A *Streptococcus*) is bacitracin-sensitive, making bacitracin testing useful for its identification.

Why Not Other Options?

A. *Bordetella pertussis* ❌

Incorrect. *B. pertussis* is identified by culture on Bordet-Gengou or Regan-Lowe medium, PCR, or antigen detection, not by optochin sensitivity.

B. *Bacillus anthracis* ❌

Incorrect. *B. anthracis* is identified by its large Gram-positive rods, capsule, non-motility, and characteristic colony morphology rather than optochin sensitivity.

D. *Mycobacterium tuberculosis* ❌

Incorrect. *M. tuberculosis* is identified by acid-fast staining, culture on Lowenstein-Jensen medium, and molecular tests such as PCR.

Key exam point

✦ Bacitracin-sensitive → *S. pyogenes* (Group A Streptococcus)

✦ Optochin-sensitive → *S. pneumoniae*

✦ *S. pneumoniae* is also bile soluble, unlike viridans streptococci.

✦ Both *S. pneumoniae* and viridans streptococci are alpha-hemolytic, but only *S. pneumoniae* is optochin-sensitive.

Remember

✓ Bacitracin → *S. pyogenes*

✓ Optochin + Bile solubility → *S. pneumoniae*

3. Otitis media is caused by all except one:

A. *Mycobacterium tuberculosis*

B. *Streptococcus pneumoniae*

C. *Moraxella catarrhalis*

D. Nontypeable *Haemophilus influenzae*

Answer A. *Mycobacterium tuberculosis*

Otitis media is an infection of the middle ear and is especially common in children due to the shorter and more horizontal Eustachian tube. The most common bacterial causes are *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Moraxella catarrhalis*. Although *Mycobacterium tuberculosis* can rarely involve the middle ear, it is not a common cause of acute otitis media and is therefore the exception.

Why Not Other Options?

B. *Streptococcus pneumoniae* ❌

Incorrect. It is the most common bacterial cause of acute otitis media.

C. *Moraxella catarrhalis* ❌

Incorrect. It is a frequent cause of otitis media, particularly in children.

D. Nontypeable *Haemophilus influenzae* ❌

Incorrect. It is one of the major causes of acute otitis media and is commonly associated with recurrent infections.

Key exam point

💡 The three most common bacterial causes of acute otitis media are:

👉 *Streptococcus pneumoniae*

👉 Nontypeable *Haemophilus influenzae*

👉 *Moraxella catarrhalis*

✦ Risk factors include young age, upper respiratory tract infections, daycare attendance, and Eustachian tube dysfunction.

✦ *Mycobacterium tuberculosis* is a rare cause of chronic ear infection, not a typical cause of acute otitis media.

Remember

✓ Acute otitis media = *S. pneumoniae* + *H. influenzae* + *M. catarrhalis*

✓ Tuberculosis is not a common cause of otitis media.

4. Mycolic acid is found in the cell wall of:

A. *Bordetella pertussis*

B. *Bacillus anthracis*

C. *Streptococcus pneumoniae*

D. *Mycobacterium tuberculosis*

Answer D. *Mycobacterium tuberculosis*

Mycolic acid is a long-chain fatty acid that is a major component of the cell wall of *Mycobacterium tuberculosis*. It gives the bacterium a waxy, lipid-rich cell wall, making it resistant to drying, many disinfectants, and some antibiotics. The presence of mycolic acid is also responsible for the acid-fast property of *M. tuberculosis*, allowing it to retain the primary stain during Ziehl-Neelsen acid-fast staining even after acid-alcohol decolorization.

Why Not Other Options?

A. *Bordetella pertussis* ✗

Incorrect. *B. pertussis* is a Gram-negative coccobacillus that causes whooping cough (pertussis). Its cell wall contains lipopolysaccharide (LPS) like other Gram-negative bacteria but does not contain mycolic acid.

B. *Bacillus anthracis* ❌

Incorrect. *B. anthracis* is a large Gram-positive, spore-forming bacillus that causes anthrax. Its cell wall is composed mainly of thick peptidoglycan, and it lacks mycolic acid.

C. *Streptococcus pneumoniae* ❌

Incorrect. *S. pneumoniae* is a Gram-positive, encapsulated diplococcus that commonly causes pneumonia, meningitis, otitis media, and sinusitis. Its cell wall contains peptidoglycan and teichoic acids, but not mycolic acid.

Key exam point

💡 Mycolic acid is the hallmark of *Mycobacterium* species.

💡 It gives the cell wall a waxy appearance and makes the bacteria acid-fast.

💡 *Mycobacterium tuberculosis* is identified by Ziehl-Neelsen acid-fast staining.

💡 The lipid-rich cell wall contributes to resistance to drying, disinfectants, and many antibiotics.

Remember

- ✓ Mycolic acid → Mycobacterium tuberculosis
- ✓ Acid-fast bacillus = Mycolic acid in the cell wall

5. Which is not a typical clinical presentation of pulmonary tuberculosis?

- A. Pharyngitis
- B. Night sweating
- C. Haemoptysis (bloody sputum)
- D. Productive cough

Answer A. Pharyngitis

Pulmonary tuberculosis (TB) is a chronic infectious disease caused by Mycobacterium tuberculosis that primarily affects the lungs. Patients typically present with a persistent productive cough lasting more than 2–3 weeks, hemoptysis, night sweats, fever, weight loss, and fatigue. Pharyngitis (sore throat) is not a typical feature of pulmonary TB and is more commonly associated with viral or bacterial infections of the upper respiratory tract.

Why Not Other Options?

B. Night sweating ❌

Incorrect. Night sweats are one of the classic constitutional symptoms of pulmonary tuberculosis and are commonly associated with prolonged fever and weight loss.

C. Haemoptysis (bloody sputum) ❌

Incorrect. Hemoptysis is a common manifestation of pulmonary TB, especially in advanced disease, due to destruction of lung tissue and blood vessels.

D. Productive cough ❌

Incorrect. A chronic productive cough is the most common symptom of pulmonary TB. The sputum may be mucoid, purulent, or blood-stained.

Key exam point

✦ Pulmonary TB is caused by *Mycobacterium tuberculosis*.

✦ Common symptoms: Persistent productive cough, hemoptysis, fever, night sweats, weight loss, and fatigue.

✦ Diagnosis is confirmed by acid-fast staining, culture, GeneXpert (PCR), and chest X-ray.

✦ TB spreads through airborne droplets from infected individuals.

Remember

✓ Pulmonary TB = Cough + Hemoptysis + Night sweats + Weight loss

✓ Pharyngitis is not a typical presentation of pulmonary tuberculosis.

6. Which form could be used as a biological weapon?

A. Cutaneous anthrax

B. Intestinal anthrax

C. Pulmonary anthrax

D. Cerebral anthrax

Answer C. Pulmonary anthrax

Pulmonary (inhalational) anthrax is the form of anthrax that has been associated with biological warfare and bioterrorism. It occurs when spores of *Bacillus anthracis* are inhaled into the lungs. The spores are taken up by alveolar macrophages, transported to mediastinal lymph nodes, and germinate into vegetative bacteria that produce potent toxins. This form is highly lethal if not diagnosed and treated promptly, making it the form of greatest concern in bioterrorism.

Why Not Other Options?

A. Cutaneous anthrax ❌

Incorrect. Cutaneous anthrax is the most common form of anthrax and occurs when spores enter through a break in the skin. It produces a characteristic painless black eschar and usually has a good prognosis with appropriate antibiotic treatment.

B. Intestinal anthrax ❌

Incorrect. Intestinal (gastrointestinal) anthrax results from eating contaminated or undercooked meat from infected animals. It causes severe abdominal pain, vomiting, and bloody diarrhea but is not the form typically used as a biological weapon.

D. Cerebral anthrax ❌

Incorrect. Cerebral anthrax is rare and usually occurs as a complication of severe systemic anthrax infection, such as anthrax meningitis. It is not a primary form associated with biological weapon use.

Key exam point

💡 *Bacillus anthracis* is a Gram-positive, spore-forming bacillus.

✦ Pulmonary (inhalational) anthrax is the form associated with bioterrorism.

✦ Virulence is due to protective antigen, edema factor, and lethal factor.

✦ Cutaneous anthrax is the most common form and presents with a painless black eschar.

Remember

✓ Biological weapon → Pulmonary (inhalational) anthrax

✓ Black eschar → Cutaneous anthrax

✓ Spore-forming bacterium → *Bacillus anthracis*

7. Paroxysmal stage of pertussis is characterized by:

A. Destruction or removal of ciliated cells

B. Common cold-like symptoms

C. Rarely occurring coughing

D. Is a convalescent stage

Answer A. Destruction or removal of ciliated cells

Pertussis (whooping cough), caused by *Bordetella pertussis*, progresses through three clinical stages: the catarrhal stage, paroxysmal stage, and convalescent stage. During the paroxysmal stage, the bacteria have already damaged the ciliated epithelial cells of the respiratory tract through the action of pertussis toxin and tracheal cytotoxin. This damage impairs mucociliary clearance, leading to repeated episodes of

severe coughing (paroxysms) followed by the characteristic high-pitched inspiratory "whoop" and sometimes post-tussive vomiting.

Why Not Other Options?

B. Common cold-like symptoms ❌

Incorrect. These symptoms occur during the catarrhal stage, the initial phase of pertussis, and include runny nose, sneezing, mild fever, and a mild cough.

C. Rarely occurring coughing ❌

Incorrect. The paroxysmal stage is characterized by frequent, severe coughing fits, not rare coughing. These coughing spells may occur many times a day and are often worse at night.

D. Is a convalescent stage ❌

Incorrect. The convalescent stage is the recovery phase, during which coughing gradually becomes less frequent and less severe over weeks to months.

Key exam point

💡 Pertussis is caused by *Bordetella pertussis*.

💡 Catarrhal stage: Common cold-like symptoms; patient is most infectious.

💡 Paroxysmal stage: Severe coughing fits, inspiratory "whoop," post-tussive vomiting, and damage to ciliated respiratory epithelium.

✦✦ Convalescent stage: Gradual recovery with decreasing cough frequency.

Remember

- ✓ Catarrhal = Cold-like symptoms
- ✓ Paroxysmal = Whooping cough + Ciliated cell damage
- ✓ Convalescent = Recovery phase

8. Pontiac fever is caused by:

- A. *Streptococcus pyogenes*
- B. *Haemophilus influenzae*
- C. *Legionella pneumophila*
- D. *Corynebacterium diphtheriae*

Answer C. *Legionella pneumophila*

Pontiac fever is a mild, self-limiting respiratory illness caused by *Legionella pneumophila*, the same organism responsible for Legionnaires' disease. It is transmitted through inhalation of contaminated aerosols, commonly from sources such as water systems, cooling towers, and air-conditioning systems. Unlike Legionnaires' disease, Pontiac fever does not usually cause pneumonia and presents mainly with fever, headache, muscle aches, and flu-like symptoms.

Why Not Other Options?

A. *Streptococcus pyogenes* ✗

Incorrect. *S. pyogenes* causes infections such as pharyngitis, scarlet fever, cellulitis, and rheumatic fever, but it is not associated with Pontiac fever.

B. *Haemophilus influenzae* ❌

Incorrect. *H. influenzae* can cause pneumonia, meningitis, otitis media, and epiglottitis, especially in susceptible individuals, but it does not cause Pontiac fever.

D. *Corynebacterium diphtheriae* ❌

Incorrect. *C. diphtheriae* causes diphtheria, characterized by an upper respiratory infection with a gray pseudomembrane and toxin-mediated complications.

Key exam point

💡 *Legionella pneumophila* causes two main clinical syndromes:

💡 Legionnaires' disease: Severe atypical pneumonia with systemic symptoms.

💡 Pontiac fever: Mild, flu-like illness without pneumonia.

💡 Transmission occurs through inhalation of contaminated water aerosols.

Remember

✓ Pontiac fever → *Legionella pneumophila*

✓ *Legionella* = Water source + Atypical pneumonia (Legionnaires') or flu-like illness (Pontiac fever)

9. Which statement is not true about antigenic shift?

- A. Occurs in Influenza A and B
- B. Occurs only in Influenza A
- C. There is formation/appearance of a new viral strain
- D. There is genome exchange of two strains

Answer A. Occurs in Influenza A and B

Antigenic shift is a major, abrupt change in the influenza virus caused by reassortment (genome exchange) between two different Influenza A viruses infecting the same host cell. This results in the emergence of a new influenza A subtype with novel hemagglutinin (HA) and/or neuraminidase (NA) antigens. Because the population has little or no pre-existing immunity, antigenic shift can lead to influenza pandemics. Influenza B does not undergo antigenic shift because it has no significant animal reservoir for reassortment.

Why Not Other Options?

B. Occurs only in Influenza A ❌

Incorrect. This is true. Antigenic shift occurs only in Influenza A, not in Influenza B or C.

C. There is formation/appearance of a new viral strain ❌

Incorrect. This is true. Antigenic shift produces a new viral strain or subtype, which may spread rapidly through the population.

D. There is genome exchange of two strains ❌

Incorrect. This is true. Antigenic shift occurs through genetic reassortment of the segmented RNA genomes of two different Influenza A viruses.

Key exam point

✦✦ Antigenic shift occurs only in Influenza A.

✦✦ It results from genome reassortment between two different viral strains.

✦✦ It causes the emergence of new influenza subtypes.

✦✦ Antigenic shift can cause pandemics, whereas antigenic drift causes seasonal epidemics.

Remember

✓ Shift = Sudden = Influenza A only = Pandemic

✓ Drift = Gradual mutations = Influenza A & B = Seasonal epidemics

10. Virus that is transmitted commonly through inhalation of infected rodent excrement and saliva:

A. Hantavirus

B. Haemophilus influenzae

C. Orthomyxoviruses

D. Corynebacterium diphtheriae

Answer A. Hantavirus

Hantavirus is a zoonotic virus transmitted mainly by inhalation of aerosolized urine, feces, saliva, or nesting materials of infected rodents. Humans become infected when they breathe in contaminated dust, especially while cleaning rodent-infested areas. Depending on the virus species, hantavirus can cause Hantavirus Pulmonary Syndrome (HPS) or Hemorrhagic Fever with Renal Syndrome (HFRS). Human-to-human transmission is rare for most hantaviruses.

Why Not Other Options?

B. Haemophilus influenzae ❌

Incorrect. H. influenzae is a Gram-negative bacterium, not a virus. It spreads through respiratory droplets and causes diseases such as otitis media, pneumonia, meningitis, and epiglottitis.

C. Orthomyxoviruses ❌

Incorrect. Orthomyxoviruses are the influenza viruses, which are transmitted mainly through respiratory droplets and aerosols from infected people, not through rodent excreta.

D. Corynebacterium diphtheriae ❌

Incorrect. C. diphtheriae is a Gram-positive bacterium transmitted by respiratory droplets or direct contact. It causes diphtheria, not rodent-borne disease.

Key exam point

💡 Hantavirus is transmitted by inhalation of aerosolized rodent urine, feces, and saliva.

💡 It is a zoonotic virus, with rodents serving as the natural reservoir.

✦ It can cause Hantavirus Pulmonary Syndrome (HPS) or Hemorrhagic Fever with Renal Syndrome (HFRS).

✦ Prevention focuses on avoiding exposure to rodent-infested environments and using proper protective measures during cleanup.

Remember

✓ Hantavirus → Rodents → Inhalation of contaminated excreta

✓ Rodents are the reservoir; humans are infected by aerosolized rodent waste.

11. Formation of a fungal ball in a cavity formed in pre-existing lung disease is commonly seen in infection with:

A. *Blastomyces dermatitidis*

B. *Histoplasma capsulatum*

C. *Coccidioides immitis*

D. *Aspergillus fumigatus*

Answer D. *Aspergillus fumigatus*

Aspergillus fumigatus commonly forms an aspergilloma (fungal ball) within pre-existing lung cavities, such as those caused by tuberculosis, sarcoidosis, bronchiectasis, or lung abscesses. The fungal ball consists of fungal hyphae, inflammatory cells, mucus, and debris. Patients may be asymptomatic or present with hemoptysis, which can sometimes be severe or life-threatening.

Why Not Other Options?

A. *Blastomyces dermatitidis* ❌

Incorrect. *Blastomyces* causes blastomycosis, a systemic fungal infection that primarily affects the lungs and skin. It does not typically form fungal balls in pre-existing lung cavities.

B. *Histoplasma capsulatum* ❌

Incorrect. *Histoplasma* causes histoplasmosis, usually acquired by inhaling spores from bird or bat droppings. It may produce pulmonary nodules or chronic lung disease but does not commonly form aspergillomas.

C. *Coccidioides immitis* ❌

Incorrect. *Coccidioides* causes coccidioidomycosis (Valley fever), which may produce pulmonary cavities in some patients, but fungal ball formation is classically associated with *Aspergillus fumigatus*.

Key exam point

💡 *Aspergillus fumigatus* commonly causes aspergilloma (fungal ball) in pre-existing lung cavities.

💡 Common predisposing conditions include tuberculosis, bronchiectasis, sarcoidosis, and lung abscesses.

💡 Hemoptysis is the most characteristic clinical manifestation and may be severe.

💡 *Aspergillus* can also cause allergic bronchopulmonary aspergillosis (ABPA) and invasive aspergillosis in immunocompromised patients.

Remember

- ✓ Fungal ball (aspergilloma) → *Aspergillus fumigatus*
- ✓ Pre-existing lung cavity + Hemoptysis = Think Aspergilloma

12. Croup is caused by:

- A. Parainfluenza virus
- B. Coxsackie A virus
- C. Rhinoviruses
- D. Coronaviruses

Answer A. Parainfluenza virus

Croup (laryngotracheobronchitis) is an acute viral infection of the larynx, trachea, and bronchi, occurring mainly in young children. The most common cause is the Parainfluenza virus (especially type 1). Inflammation and edema of the upper airway produce the characteristic barking cough, hoarseness, inspiratory stridor, and varying degrees of respiratory distress. Symptoms are often preceded by a mild upper respiratory tract infection.

Why Not Other Options?

B. Coxsackie A virus ❌

Incorrect. Coxsackie A virus commonly causes herpangina and hand, foot, and mouth disease (HFMD). It is not the usual cause of croup.

C. Rhinoviruses ❌

Incorrect. Rhinoviruses are the most common cause of the common cold. Although they infect the upper respiratory tract, they are not the primary cause of croup.

D. Coronaviruses ❌

Incorrect. Coronaviruses mainly cause upper respiratory tract infections, and some strains can cause severe lower respiratory disease. However, Parainfluenza virus is the classic and most common cause of croup.

Key exam point

💡 Parainfluenza virus type 1 is the most common cause of croup.

💡 Croup is also known as laryngotracheobronchitis.

💡 Typical features include barking cough, inspiratory stridor, hoarseness, and respiratory distress.

💡 It mainly affects children between 6 months and 3 years of age.

Remember

✓ Croup = Parainfluenza virus

✓ Barking cough + Inspiratory stridor + Hoarseness = Croup

13. Herpangina is caused by a virus called:

A. Parainfluenza virus

B. Coxsackie A virus

C. Rhinoviruses

D. Coronaviruses

Answer B. Coxsackie A virus

Herpangina is an acute viral illness caused primarily by Coxsackie A virus, a member of the Enterovirus genus. It commonly affects children, especially during the summer and early autumn. The disease is characterized by fever, sore throat, painful swallowing (odynophagia), and small vesicles or ulcers on the soft palate, tonsils, uvula, and posterior pharynx. It is usually self-limiting, resolving within a week.

Why Not Other Options?

A. Parainfluenza virus ❌

Incorrect. Parainfluenza virus is the most common cause of croup (laryngotracheobronchitis) and is not associated with herpangina.

C. Rhinoviruses ❌

Incorrect. Rhinoviruses are the most common cause of the common cold, causing symptoms such as runny nose, sneezing, and sore throat, but they do not typically cause herpangina.

D. Coronaviruses ❌

Incorrect. Coronaviruses primarily cause upper respiratory tract infections and, in some cases, severe lower respiratory disease. They are not the usual cause of herpangina.

Key exam point

💡 Herpangina is most commonly caused by Coxsackie A virus.

✦ It presents with fever, sore throat, and painful vesicles or ulcers in the posterior oral cavity.

✦ It mainly affects children and is spread by the fecal-oral route and respiratory secretions.

✦ It is usually a self-limiting illness requiring supportive treatment.

Remember

✓ Herpangina → Coxsackie A virus

✓ Posterior mouth ulcers + Fever = Herpangina

✓ Coxsackie A also causes Hand, Foot, and Mouth Disease (HFMD)

14. Which bacterium is found in high concentration in the placenta of farm animals?

A. *Streptococcus pyogenes*

B. *Haemophilus influenzae*

C. *Coxiella burnetii*

D. *Corynebacterium diphtheriae*

Answer C. *Coxiella burnetii*

Coxiella burnetii is the causative agent of Q fever. It is found in very high concentrations in the placenta, amniotic fluid, and birth products of infected farm animals, especially sheep, goats, and cattle. Humans are usually infected by inhaling aerosolized particles contaminated with these materials during animal births. Because the organism is highly infectious, even a small number of bacteria can cause disease.

Why Not Other Options?

A. *Streptococcus pyogenes* ❌

Incorrect. *S. pyogenes* is a Group A β -hemolytic *Streptococcus* that commonly causes pharyngitis, impetigo, cellulitis, and rheumatic fever. It is not associated with the placenta of farm animals.

B. *Haemophilus influenzae* ❌

Incorrect. *H. influenzae* is a Gram-negative coccobacillus that colonizes the human respiratory tract and causes diseases such as otitis media, pneumonia, meningitis, and epiglottitis. It is not transmitted from farm animals.

D. *Corynebacterium diphtheriae* ❌

Incorrect. *C. diphtheriae* causes diphtheria, a human upper respiratory tract infection characterized by a gray pseudomembrane. It is spread by respiratory droplets, not through infected animal placentas.

Key exam point

💡 Q fever is caused by *Coxiella burnetii*.

💡 The organism is present in high concentrations in the placenta, amniotic fluid, and birth products of infected sheep, goats, and cattle.

💡 Humans are infected mainly by inhalation of contaminated aerosols, not by animal bites.

💡 Q fever commonly presents with fever, headache, atypical pneumonia, or hepatitis.

Remember

✓ Placenta of farm animals → *Coxiella burnetii*

✓ Q fever = Sheep, goats, cattle + Inhalation of contaminated birth products

15. Tonsillitis or pharyngitis is commonly caused by:

A. Pertussis (*Bordetella pertussis*)

B. *Streptococcus pyogenes*

C. *Streptococcus pneumoniae*

D. *Mycobacterium tuberculosis*

Answer B. *Streptococcus pyogenes*

Streptococcus pyogenes (Group A β -hemolytic *Streptococcus*) is the most common bacterial cause of acute pharyngitis and tonsillitis. It is spread through respiratory droplets and typically presents with sore throat, fever, painful swallowing, swollen tonsils with exudates, and tender anterior cervical lymph nodes. Early diagnosis and treatment are important to prevent complications such as acute rheumatic fever and post-streptococcal glomerulonephritis.

Why Not Other Options?

A. Pertussis (*Bordetella pertussis*) ✗

Incorrect. B. pertussis causes whooping cough (pertussis), characterized by paroxysmal coughing fits with an inspiratory "whoop," rather than tonsillitis or pharyngitis.

C. *Streptococcus pneumoniae* ✗

Incorrect. *S. pneumoniae* commonly causes pneumonia, otitis media, sinusitis, and meningitis. It is not a common cause of acute pharyngitis or tonsillitis.

D. *Mycobacterium tuberculosis* ❌

Incorrect. *M. tuberculosis* primarily causes pulmonary tuberculosis. Although it can rarely involve the oropharynx, it is not a common cause of acute tonsillitis or pharyngitis.

Key exam point

💡 The most common bacterial cause of acute pharyngitis is *Streptococcus pyogenes*.

💡 It is a Group A β -hemolytic *Streptococcus* (GAS).

💡 Clinical features include fever, sore throat, tonsillar exudates, and tender anterior cervical lymph nodes.

💡 Untreated infection may lead to acute rheumatic fever and post-streptococcal glomerulonephritis.

Remember

✓ Pharyngitis/Tonsillitis → *Streptococcus pyogenes*

✓ Group A Strep = Sore throat + Tonsillar exudates + Rheumatic fever risk

Exam:-Microbiology

Year:-2015 E.C

Dr. Doc

1. Which of the following is TRUE regarding the development of pneumonia?

- A. Pneumonia is an acute inflammatory condition within the parenchyma of the lung.
- B. Immunocompromised individuals are at increased risk of developing pneumonia.
- C. Pneumonia often develops as a consequence of bacterial colonization and microaspiration of upper respiratory tract secretions during impaired pulmonary defense mechanisms.
- D. All of the above.

Answer: D. All of the above

Pneumonia is an acute infection and inflammatory condition of the lung parenchyma, particularly involving the alveoli. People with weakened immune systems are more susceptible because their pulmonary defense mechanisms cannot effectively eliminate invading microorganisms. Pneumonia can also develop when microorganisms from the upper respiratory tract are microaspirated into the lower respiratory tract, especially when normal protective mechanisms such as the cough reflex and mucociliary clearance are impaired. Therefore, all three statements are correct.

Why Not Other Options?

A. Pneumonia is an acute inflammatory condition within the parenchyma of the lung. ✖

This statement is true, but it is not the best answer because B and C are also correct.

B.

Immunocompromised individuals are at increased risk. ✖

This is true. Reduced immune function makes it easier for pathogens to invade and multiply in the lungs.

C. Microaspiration can cause pneumonia. ✖

This is also true. Aspiration of colonized upper respiratory secretions is an important mechanism of pneumonia, particularly when pulmonary defense mechanisms are impaired.

Key exam point

✦ Pneumonia involves infection and inflammation of the lung parenchyma.

✦ Impaired immunity and impaired pulmonary defenses increase the risk.

✦ Microaspiration of upper respiratory secretions is an important route of infection.

Remember

✓ Pneumonia → lung parenchyma

✓ Aspiration + impaired defense → increased pneumonia risk

2. A 7-year-old boy has swollen and erythematous pharynx and tonsils, tonsillar exudate, fever, tender cervical lymphadenopathy, and leukocytosis. What is the most likely diagnosis?

A. Acute viral infection

B. Acute bacterial pharyngitis

C. Severe acute respiratory syndrome

D. Acute lower respiratory infection

Answer: B. Acute bacterial pharyngitis

The findings strongly suggest acute bacterial pharyngitis, particularly pharyngitis caused by *Streptococcus pyogenes* (Group A *Streptococcus*). Typical findings include fever, tonsillar erythema and swelling, tonsillar exudate, and tender cervical lymphadenopathy. Leukocytosis also supports a bacterial infection. In contrast, viral pharyngitis commonly has symptoms such as cough, runny nose, conjunctivitis, or hoarseness.

Why Not Other Options?

A. Acute viral infection ❌

Viral infections can cause sore throat and fever, but they are more commonly associated with cough, rhinorrhea, conjunctivitis, or hoarseness.

C. Severe acute respiratory syndrome ❌

SARS primarily affects the lower respiratory tract, causing symptoms such as fever, cough, and difficulty breathing rather than classic exudative tonsillitis.

D. Acute lower respiratory infection ❌

The main findings are localized to the pharynx and tonsils, with no evidence of pneumonia or another lower respiratory infection.

Key exam point

💡 Fever + tonsillar exudate + tender cervical lymph nodes → strongly suggests *S. pyogenes* pharyngitis.

Remember

✓ *S. pyogenes* → Group A Streptococcus → bacterial pharyngitis

✓ Exudative tonsils + tender cervical nodes = important clue

3. Which statement is NOT true about *Corynebacterium diphtheriae*?

- A. Causes lower respiratory infection
- B. Causes upper respiratory infection
- C. Secretes an exotoxin that inhibits protein synthesis
- D. Forms pseudomembrane on the pharynx

Answer: A. Causes lower respiratory infection

Corynebacterium diphtheriae typically causes an upper respiratory tract infection, especially involving the pharynx and tonsils. The organism produces diphtheria exotoxin, which inhibits protein synthesis by inactivating elongation factor-2 (EF-2). A characteristic grayish-white pseudomembrane can form over the pharynx and tonsils.

Why Not Other Options?

B. Causes upper respiratory infection ❌

This is true. Respiratory diphtheria primarily affects the pharynx, tonsils, and upper respiratory tract.

C. Secretes an exotoxin that inhibits protein synthesis ❌

This is true. Diphtheria toxin inactivates EF-2, preventing protein synthesis and causing cell death.

D. Forms pseudomembrane on the pharynx ❌

This is true. The adherent pseudomembrane is one of the classic features of respiratory diphtheria.

Key exam point

💡 Diphtheria toxin → EF-2 inhibition → inhibition of protein synthesis.

💡 Pharyngeal pseudomembrane → classic feature of respiratory diphtheria.

Remember

✓ Diphtheria → upper respiratory infection

✓ Pseudomembrane + exotoxin = classic clues

4. Paroxysmal stage of pertussis is characterized by:

A. Common cold-like symptoms

B. Rarely occurring coughing

C. Convalescent stage

D. Repetitive cough with whoops

Answer: D. Repetitive cough with whoops

The paroxysmal stage is characterized by repeated, severe bursts of coughing. The patient may have multiple rapid coughs during a single episode, followed by a forceful inspiration that produces the characteristic "whoop." Vomiting after coughing can also occur. This stage follows the catarrhal stage, which resembles a common cold.

Why Not Other Options?

A. Common cold-like symptoms ❌

These are characteristic of the catarrhal stage, not the paroxysmal stage.

B. Rarely occurring coughing ❌

Incorrect. Coughing is frequent and occurs in severe repetitive attacks during the paroxysmal stage.

C. Convalescent stage ❌

The convalescent stage occurs later, when the frequency and severity of the cough gradually decrease.

Key exam point

💡 Paroxysmal stage → repeated coughing attacks + inspiratory whoop.

Remember

✓ Catarrhal → cold-like symptoms

✓ Paroxysmal → whooping/repetitive cough

✓ Convalescent → gradual recovery

5. A 2-day-old premature baby develops pneumonia. The mother had symptoms of UTI after delivery, and beta-hemolytic colonies were found. Which pathogen can spread from mother to baby and cause pneumonia?

A. *Streptococcus pyogenes*

B. *Streptococcus bovis*

C. *Streptococcus agalactiae*

D. *Enterococcus faecalis*

Answer: C. *Streptococcus agalactiae*

Streptococcus agalactiae, also called Group B *Streptococcus* (GBS), is an important cause of neonatal pneumonia, sepsis, and meningitis. It commonly colonizes the female genital and gastrointestinal tracts and can be transmitted to the newborn during labor or delivery. Premature infants are particularly vulnerable to severe early-onset GBS disease.

GBS is beta-hemolytic, making the culture finding an important clue in this question.

Why Not Other Options?

A. *Streptococcus pyogenes* ✗

Group A *Streptococcus* commonly causes pharyngitis, skin infections, and scarlet fever, but it is not the classic cause of early-onset neonatal pneumonia described here.

B. *Streptococcus bovis* ✗

It is more strongly associated with bacteremia and endocarditis, as well as an association with colonic disease, rather than typical neonatal pneumonia.

D. *Enterococcus faecalis* ❌

It can cause urinary tract and healthcare-associated infections, but it is not the classic maternal-to-newborn pathogen responsible for early-onset neonatal pneumonia.

Key exam point

💡 Group B Streptococcus = *S. agalactiae*.

💡 It is an important cause of neonatal pneumonia, sepsis, and meningitis.

💡 Beta-hemolysis + newborn infection + maternal colonization → think GBS.

Remember

✓ *S. agalactiae* → Group B Strep → neonatal disease

✓ *S. pyogenes* → Group A Strep → pharyngitis

6. Which bacterial pathogen is the most common cause of community-acquired pneumonia in adults?

A. *Staphylococcus aureus*

B. *Mycoplasma pneumoniae*

C. *Streptococcus pneumoniae*

D. Chlamydophila pneumoniae

Answer: C. Streptococcus pneumoniae

Streptococcus pneumoniae is the classic most common bacterial cause of community-acquired pneumonia (CAP) in adults. It is an encapsulated, Gram-positive, lancet-shaped diplococcus. It commonly causes pneumonia, otitis media, sinusitis, and meningitis.

Pneumococcal pneumonia may present with fever, productive cough, pleuritic chest pain, and lobar consolidation. Its capsule is an important virulence factor because it helps the organism resist phagocytosis.

Why Not Other Options?

A. Staphylococcus aureus ✗

It can cause severe pneumonia, particularly following influenza, but it is not the most common bacterial cause of typical community-acquired pneumonia.

B. Mycoplasma pneumoniae ✗

It is an important cause of atypical pneumonia, especially in school-aged children and young adults, but it is not the most common bacterial cause overall.

D. Chlamydophila pneumoniae ✗

It can cause atypical community-acquired pneumonia but is less common than S. pneumoniae.

Key exam point

💡 Most common classic bacterial cause of CAP → S. pneumoniae.

✦ It is an encapsulated Gram-positive diplococcus.

✦ It commonly causes pneumococcal/lobar pneumonia.

Remember

✓ Community-acquired pneumonia → *S. pneumoniae*

✓ Encapsulated + Gram-positive + diplococcus → *S. pneumoniae*

7. Which of the following bacteria is an obligate aerobe and a highly contagious pathogen that may cause serious respiratory infections in young children?

A. *Haemophilus ducreyi*

B. *Francisella tularensis*

C. *Haemophilus parainfluenzae*

D. *Bordetella pertussis*

Answer: D. *Bordetella pertussis*

Bordetella pertussis is a small Gram-negative coccobacillus and an important cause of pertussis (whooping cough), particularly in infants and young children. It is highly contagious and spreads through respiratory droplets. The organism produces several virulence factors, including pertussis toxin, which contributes to lymphocytosis and disease manifestations.

Pertussis classically progresses through catarrhal, paroxysmal, and convalescent stages. The paroxysmal stage is characterized by repeated severe coughing followed by an inspiratory "whoop."

Why Not Other Options?

A. *Haemophilus ducreyi* ❌

Causes chancroid, a sexually transmitted infection characterized by painful genital ulcers. It is not a major respiratory pathogen.

B. *Francisella tularensis* ❌

Causes tularemia. It is highly infectious but is associated with different clinical forms such as ulceroglandular and pneumonic tularemia rather than classic childhood whooping cough.

C. *Haemophilus parainfluenzae* ❌

It is part of the normal respiratory flora and can occasionally cause respiratory or invasive infections, but it is not the classic highly contagious cause of severe childhood coughing illness.

Key exam point

💡 *Bordetella pertussis* → whooping cough.

💡 It is a Gram-negative coccobacillus transmitted by respiratory droplets.

💡 Produces pertussis toxin and causes characteristic paroxysmal coughing.

Remember

✓ B. pertussis → pertussis → whooping cough

✓ Catarrhal → Paroxysmal → Convalescent

8. Which bacterium is a strict aerobic, Gram-negative, rod-shaped organism associated with cystic fibrosis, burns, hospital-acquired infections, ventilator-associated pneumonia, and sepsis?

- A. *Pseudomonas aeruginosa*
- B. *Klebsiella pneumoniae*
- C. *Helicobacter pylori*
- D. *Campylobacter jejuni*

Answer: A. *Pseudomonas aeruginosa*

Pseudomonas aeruginosa is a Gram-negative, aerobic, rod-shaped bacterium and an important opportunistic pathogen. It is particularly associated with patients with cystic fibrosis, severe burns, immunosuppression, and prolonged hospitalization.

It is a major cause of hospital-acquired pneumonia, ventilator-associated pneumonia, urinary tract infections, wound infections, bacteremia, and sepsis. It can form a biofilm, especially in the lungs of patients with cystic fibrosis, making infections difficult to eradicate.

Why Not Other Options?

B. *Klebsiella pneumoniae* ✗

It is also a Gram-negative rod and can cause pneumonia, particularly in hospitalized or debilitated patients. However, it is typically encapsulated and nonmotile, and it is not the classic organism associated with cystic fibrosis.

C. *Helicobacter pylori* ✗

This organism colonizes the stomach and is associated with peptic ulcers and gastric cancer. It is microaerophilic, not a strict aerobe.

D. *Campylobacter jejuni* ❌

It is a microaerophilic, curved Gram-negative rod and is an important cause of bacterial gastroenteritis, not the classic hospital-acquired respiratory pathogen.

Key exam point

💡 *Pseudomonas aeruginosa* → opportunistic Gram-negative rod.

💡 Strong association with cystic fibrosis and hospital-acquired infections.

💡 Important cause of ventilator-associated pneumonia and sepsis.

Remember

✓ Cystic fibrosis → think *Pseudomonas*

✓ Burns + hospital infection → think *Pseudomonas*

9. Which human fungal species also occurs in the lungs of animals and causes pneumonia in immunocompromised patients?

A. *Aspergillus niger*

B. *Pneumocystis jirovecii*

C. *Penicillium marneffe*

D. *Fusarium*

Answer: B. *Pneumocystis jirovecii*

Pneumocystis jirovecii is an opportunistic fungal pathogen that primarily causes pneumonia in immunocompromised individuals, especially patients with advanced HIV/AIDS and very low CD4 counts. It causes *Pneumocystis pneumonia* (PCP), which typically presents with progressive dry cough, fever, and shortness of breath.

The organism is also closely related to *Pneumocystis* species found in animals. In humans, it primarily affects the lungs and can cause severe respiratory disease when cellular immunity is compromised.

Why Not Other Options?

A. *Aspergillus niger* ✗

Aspergillus species can cause pulmonary disease, particularly in immunocompromised patients, but the classic opportunistic pneumonia in this question is *Pneumocystis jirovecii*.

C. *Penicillium marneffe* ✗

Now called *Talaromyces marneffe*, it causes systemic infection particularly in immunocompromised patients, especially in Southeast Asia. It is not the classic organism responsible for PCP.

D. *Fusarium* ✗

Fusarium can cause serious infections in severely immunocompromised patients, but it is more commonly associated with disseminated infection and keratitis rather than classic PCP.

Key exam point

✦ Pneumocystis jirovecii → opportunistic fungal pneumonia.

✦ Particularly important in advanced HIV/AIDS and other states of impaired cellular immunity.

Remember

✓ Immunocompromised + dry cough + dyspnea → think P. jirovecii

✓ PCP = Pneumocystis pneumonia

10. A 69-year-old woman has fever, cough, chest pain, body aches, and lobar pneumonia. Round diplococci in chains were isolated. Which pathogen is most likely?

A. Streptococcus pneumoniae

B. Neisseria gonorrhoeae

C. Haemophilus influenzae

D. Bordetella pertussis

Answer: A. Streptococcus pneumoniae

The most likely pathogen is Streptococcus pneumoniae. It is a Gram-positive, lancet-shaped diplococcus, often seen in pairs or short chains. It is the classic bacterial cause of community-acquired pneumonia, especially in older adults.

Pneumococcal pneumonia commonly causes fever, productive cough, pleuritic chest pain, and lobar consolidation. The patient's age and classic lobar pneumonia presentation further support this diagnosis.

Why Not Other Options?

B. *Neisseria gonorrhoeae* ✖

This is a Gram-negative diplococcus associated primarily with sexually transmitted infections such as gonorrhea. It is not a typical cause of lobar pneumonia.

C. *Haemophilus influenzae* ✖

It can cause pneumonia, particularly in older adults and patients with underlying lung disease, but it is a small Gram-negative coccobacillus, not a diplococcus.

D. *Bordetella pertussis* ✖

Causes whooping cough and is associated with prolonged paroxysmal coughing, not classic lobar pneumonia with diplococci.

Key exam point

💡 Lobar pneumonia + Gram-positive diplococci → *S. pneumoniae*.

💡 *S. pneumoniae* is encapsulated and lancet-shaped.

Remember

✓ Pneumonia + diplococci → *S. pneumoniae*

✓ Lobar consolidation → classic pneumococcal clue

11. All of the following statements regarding *Mycoplasma pneumoniae* are true, EXCEPT:

- A. It lacks a cell wall
- B. It is a causative agent of atypical pneumonia in humans
- C. It is pleomorphic and one of the smallest free-living microorganisms
- D. It can be cultivated on regular nutrient agar media

Answer: D. It can be cultivated on regular nutrient agar media

Mycoplasma pneumoniae is a major cause of atypical pneumonia, particularly in children, adolescents, and young adults. It is unusual among bacteria because it completely lacks a cell wall and instead has a membrane containing sterols.

Because it lacks a cell wall, it is pleomorphic and is not susceptible to antibiotics that target bacterial cell-wall synthesis, such as β -lactams. It is also one of the smallest free-living organisms.

It requires specialized media for culture and therefore cannot be routinely cultivated on ordinary nutrient agar.

Why Not Other Options?

A. It lacks a cell wall ❌

True. This is the major distinguishing feature of *Mycoplasma*.

B. Causes atypical pneumonia ❌

True. *M. pneumoniae* is a classic cause of atypical ("walking") pneumonia.

C. Pleomorphic and very small ❌

True. Without a rigid cell wall, the organism can assume different shapes and is among the smallest free-living organisms.

D. Can be cultivated on regular nutrient agar ❌

False. Mycoplasma requires specialized media containing sterols and is difficult and slow to culture.

Key exam point

💡 Mycoplasma pneumoniae = no cell wall.

💡 Requires sterol-containing specialized media.

💡 Causes atypical pneumonia.

Remember

✓ No cell wall → no peptidoglycan

✓ No cell wall → β -lactams are ineffective

✓ Mycoplasma → atypical pneumonia

12. A 2-year-old unvaccinated child has dysphonia, a grayish exudate over the tonsils, larynx and pharynx, and massive tender cervical lymphadenitis. What is the most likely cause?

A. Corynebacterium diphtheriae

B. Streptococcus pneumoniae

C. *Bordetella pertussis*

D. *Haemophilus influenzae*

Answer: A. *Corynebacterium diphtheriae*

This is a classic presentation of respiratory diphtheria. The strongest clues are the child's lack of vaccination, grayish pseudomembrane, dysphonia, and massive cervical lymphadenopathy.

C. *diphtheriae* produces diphtheria toxin, which inhibits protein synthesis by inactivating EF-2. The toxin and local infection produce tissue necrosis and formation of the characteristic adherent pseudomembrane. Extensive cervical swelling can produce the characteristic "bull neck" appearance.

Why Not Other Options?

B. *Streptococcus pneumoniae* ❌

It commonly causes pneumonia, otitis media, sinusitis, and meningitis, but does not characteristically produce a gray adherent pharyngeal pseudomembrane.

C. *Bordetella pertussis* ❌

Causes whooping cough, characterized by paroxysmal coughing and an inspiratory whoop. It does not produce a characteristic pharyngeal pseudomembrane.

D. *Haemophilus influenzae* ❌

It can cause respiratory infections and epiglottitis, but the classic gray pseudomembrane + unvaccinated child + cervical swelling strongly indicates diphtheria.

Key exam point

💡 Unvaccinated child + gray pseudomembrane + cervical lymphadenopathy → diphtheria.

💡 C. diphtheriae toxin → EF-2 inhibition → ↓ protein synthesis.

Remember

✓ Gray pseudomembrane → Diphtheria

✓ Bull neck → severe respiratory diphtheria

✓ Unvaccinated child → important clue

13. A 4-year-old child develops a common cold, followed by a barking cough, persistent fever, and louder breathing sounds. Which pathogen is most likely?

A. Haemophilus influenzae

B. Rhinovirus

C. Parainfluenza virus

D. Corynebacterium diphtheriae

Answer: C. Parainfluenza virus

This is a classic presentation of croup (laryngotracheobronchitis), which is most commonly caused by parainfluenza virus, especially Parainfluenza virus type 1.

Croup commonly affects young children, usually after an upper respiratory tract infection. The characteristic barking cough results from inflammation and narrowing of the larynx and subglottic region. Swelling can produce inspiratory stridor (loud breathing sound). Symptoms often become worse at night.

Why Not Other Options?

A. *Haemophilus influenzae* ❌

H. influenzae type b can cause epiglottitis, which presents with high fever, drooling, difficulty swallowing, and respiratory distress. It does not typically produce the classic barking cough of croup.

B. Rhinovirus ❌

Rhinovirus is the most common cause of the common cold, but it is not the classic cause of croup.

D. *Corynebacterium diphtheriae* ❌

Diphtheria can cause upper airway obstruction and pseudomembrane formation, but the barking cough and typical age group strongly favor croup caused by parainfluenza.

Key exam point

💡 Barking cough + inspiratory stridor + young child → Croup.

💡 Most common cause → Parainfluenza virus.

Remember

✓ Croup → Parainfluenza virus

✓ Barking cough → Croup

✓ Stridor → upper airway narrowing

14. Which pathogen is a major cause of nosocomial/hospital-acquired pneumonia?

A. *Streptococcus pneumoniae*

B. *Escherichia coli*

C. *Pseudomonas aeruginosa*

D. *Acinetobacter baumannii*

E. *Neisseria meningitidis*

Answer: C. *Pseudomonas aeruginosa*

Pseudomonas aeruginosa is an important cause of hospital-acquired pneumonia (HAP) and particularly ventilator-associated pneumonia (VAP). It is an opportunistic Gram-negative rod that thrives in hospital environments and can infect patients with prolonged hospitalization, mechanical ventilation, severe illness, burns, or immunosuppression.

It is also important because it can develop multiple antibiotic resistance, making treatment difficult.

Why Not Other Options?

A. *Streptococcus pneumoniae* ✗

It is a major cause of community-acquired pneumonia, rather than the classic hospital-associated pathogen.

B. *E. coli* ✗

E. coli can cause hospital-acquired infections, including pneumonia, but it is less characteristic than *Pseudomonas* in this context.

D. *Acinetobacter baumannii* ❌

This is also an important hospital-acquired pathogen and can cause ventilator-associated pneumonia. However, *Pseudomonas* is the classic answer in this type of question.

E. *Neisseria meningitidis* ❌

Primarily causes meningitis and meningococemia, not typical hospital-acquired pneumonia.

Key exam point

💡 *Pseudomonas aeruginosa* → important HAP/VAP pathogen.

💡 Particularly associated with ventilated and critically ill patients.

Remember

✓ Community-acquired → *S. pneumoniae*

✓ Hospital-acquired/VAP → *Pseudomonas*

15. Which bacterium is the most commonly isolated pathogen from pharyngitis?

- A. *Streptococcus pneumoniae*
- B. *Staphylococcus aureus*
- C. *Streptococcus pyogenes*
- D. *Mycoplasma pneumoniae*

Answer: C. *Streptococcus pyogenes*

Streptococcus pyogenes (Group A *Streptococcus*) is the most important and classic bacterial cause of acute pharyngitis, particularly in school-aged children.

It commonly produces fever, sore throat, tonsillar erythema, tonsillar exudates, and tender anterior cervical lymphadenopathy. Recognition is important because appropriate treatment reduces the risk of acute rheumatic fever.

Why Not Other Options?

A. *S. pneumoniae* ✗

More commonly associated with pneumonia, otitis media, sinusitis, and meningitis.

B. *S. aureus* ✗

Can colonize the upper respiratory tract and cause various infections, but it is not the classic cause of acute pharyngitis.

D. *M. pneumoniae* ✗

Can cause pharyngitis, particularly as part of respiratory infection, but is primarily associated with atypical pneumonia.

Key exam point

💠 Classic bacterial pharyngitis → *S. pyogenes* (Group A Strep).

💠 Important complication → acute rheumatic fever.

Remember

✓ Sore throat + fever + tonsillar exudate → *S. pyogenes*

16. Because of time and technical limitations of viral cultures, laboratory diagnosis of viral pneumonia in adults usually relies on:

- A. Lung biopsy
- B. Acute and convalescent serologic testing
- C. Nasal swab for slide preparation and staining
- D. Rapid antigen detection, often using molecular techniques

Answer: D. Rapid antigen detection, often using molecular techniques

Viral culture is generally slow and technically demanding, so it is not usually the preferred method for rapid diagnosis. Modern diagnosis commonly uses molecular techniques such as PCR/NAAT, which can rapidly detect viral nucleic acids from respiratory specimens.

Respiratory samples can include nasopharyngeal or nasal swabs, depending on the suspected virus and test.

Why Not Other Options?

A. Lung biopsy ❌

Too invasive for routine diagnosis and generally unnecessary for uncomplicated viral pneumonia.

B. Acute and convalescent serology ❌

Serology can be useful for some infections, but it often requires comparison of antibody levels over time, making it less useful for rapid clinical decisions.

C. Nasal swab for slide preparation and staining ❌

Routine staining of a nasal swab is not a reliable general method for identifying respiratory viruses.

Key exam point

💡 PCR/NAAT → rapid and sensitive detection of respiratory viruses.

💡 Respiratory swabs are commonly used as specimens.

Remember

✓ Viral pneumonia → molecular testing/PCR is commonly used

✓ Culture = slower and more technically demanding

17. The difference between latent TB infection and active TB disease is that:

A. People with latent TB infection are not infectious, whereas people with active TB disease can be infectious.

B. Only active TB disease can be detected by a tuberculin skin test.

C. People with latent TB infection are infectious, whereas people with active TB disease are not.

D. Latent TB infection is curable but active TB disease is not.

Answer: A. People with latent TB infection are not infectious, whereas people with active TB disease can be infectious.

Latent TB infection (LTBI) means the person has *Mycobacterium tuberculosis* in the body, but the bacteria are being controlled by the immune system. The person has no symptoms and does not transmit TB to others. However, latent infection can reactivate later and become active TB, especially when immunity is weakened.

Active TB disease occurs when the bacteria multiply and cause tissue damage and clinical disease. Pulmonary or laryngeal active TB can be infectious because bacteria can be released into the air through coughing, speaking, or sneezing.

Why Not Other Options?

B. Only active TB disease can be detected by a tuberculin skin test. ❌

False. A tuberculin skin test (TST) can be positive in both latent TB infection and active TB disease. It indicates an immune response to TB but does not by itself distinguish latent from active disease.

C. People with latent TB infection are infectious, whereas people with active TB disease are not. ❌

This is the opposite of the correct concept. Latent TB is not infectious, while active pulmonary TB may be infectious.

D. Latent TB infection is curable but active TB disease is not. ❌

Both latent TB infection and active TB disease can be treated and cured with appropriate anti-TB therapy. Active TB generally requires a more intensive multidrug regimen.

Key exam point

✦ Latent TB → infected but asymptomatic and not infectious.

✦ Active TB → symptomatic disease and may be infectious, especially pulmonary TB.

Remember

✓ Latent = no disease, no transmission

✓ Active = disease + possible transmission

✓ TST/IGRA can detect TB infection but cannot alone distinguish latent from active TB

18. Which technology is most amenable to point-of-care testing for TB, allowing treatment decisions during the same visit?

- A. Smear microscopy
- B. GeneXpert MTB/RIF
- C. Culture
- D. Serological test

Answer: B. GeneXpert MTB/RIF

GeneXpert MTB/RIF is a molecular test that can rapidly detect *Mycobacterium tuberculosis* and determine rifampicin resistance directly from a clinical specimen.

This makes it particularly useful for rapid diagnosis and early treatment decisions. Unlike culture, which can take weeks because *M. tuberculosis* grows slowly, GeneXpert can provide results much more quickly.

Why Not Other Options?

A. Smear microscopy ❌

It is relatively rapid and inexpensive but has lower sensitivity, especially in patients with low bacterial loads, and cannot reliably identify drug resistance.

C. Culture ❌

Culture is highly important and is a reference method, but *M. tuberculosis* grows slowly, so results may take weeks.

D. Serological test ❌

Serologic antibody tests are not sufficiently reliable for diagnosing active TB and are not recommended as the primary diagnostic approach.

Key exam point

💡 GeneXpert MTB/RIF → rapid molecular detection of TB + rifampicin resistance.

Remember

✓ Rapid TB diagnosis → GeneXpert

✓ Culture → slow but important reference method

✓ Smear → rapid but less sensitive

19. Which form of anthrax could be used as a biological weapon?

A. Cutaneous anthrax

B. Intestinal anthrax

C. Pulmonary anthrax

D. Cerebral anthrax

Answer: C. Pulmonary anthrax

Pulmonary (inhalational) anthrax is the form associated with inhalation of *Bacillus anthracis* spores and has historically been of particular concern in biological warfare/bioterrorism because the spores can be aerosolized.

After inhalation, spores can reach the lungs and then be transported to mediastinal lymph nodes, where they germinate and produce toxin. Severe disease can progress rapidly to hemorrhagic mediastinitis, sepsis, and respiratory failure.

Why Not Other Options?

A. Cutaneous anthrax ❌

The most common form of naturally acquired anthrax. It typically produces a painless papule → vesicle → black eschar with surrounding edema.

B. Intestinal anthrax ❌

Occurs after ingestion of contaminated food and can cause severe abdominal pain, vomiting, diarrhea, and systemic disease.

D. Cerebral anthrax ❌

Not one of the main classic clinical forms of anthrax.

Key exam point

💡 Inhalational/pulmonary anthrax → associated with aerosolized spores and biological warfare concern.

💡 B. anthracis is a Gram-positive, spore-forming, encapsulated rod.

Remember

✓ Cutaneous → black eschar

✓ Inhalational → mediastinal involvement

✓ Ingestion → intestinal anthrax

20. Your patient is diagnosed with a latent tuberculosis infection. Select one correct statement that reflects this condition:

- A. The patient will feel ill and will have signs and symptoms.
- B. The patient will have a positive tuberculin skin test or IGRA test.
- C. The patient will have an abnormal chest X-ray.
- D. The patient's sputum will test positive for *Mycobacterium tuberculosis*.

Answer: B. The patient will have a positive tuberculin skin test or IGRA test.

Latent tuberculosis infection means the person has been infected with *Mycobacterium tuberculosis*, but the immune system is keeping the bacteria under control. The organisms are not causing active disease, so the patient usually has no symptoms and cannot transmit TB to others. However, the immune system has been sensitized to TB antigens, which can be detected by a tuberculin skin test (TST) or IGRA.

A person with latent TB can potentially develop active TB later if immune defenses become weakened, which is why treatment of latent infection is important.

Why Not Other Options?

A. The patient will feel ill and will have signs and symptoms. ❌

Latent TB is generally asymptomatic. Symptoms such as persistent cough, fever, night sweats, weight loss, and fatigue are more suggestive of active TB disease.

C. The patient will have an abnormal chest X-ray. ❌

A person with uncomplicated latent TB generally has a normal chest X-ray. Abnormal findings may indicate active disease or previous pulmonary TB.

D. The patient's sputum will test positive for *Mycobacterium tuberculosis*. ❌

Sputum positivity indicates active pulmonary TB with organisms being released from the respiratory tract. Latent TB does not normally produce positive sputum cultures.

Key exam point

💡 Latent TB → positive TST/IGRA, no symptoms, usually normal chest X-ray, and not infectious.

💡 The bacteria are contained, not actively causing disease.

Remember

✓ Latent = infected but inactive

✓ Active = disease + possible transmission

✓ Latent TB can reactivate later

21. Which of the following are interventions to improve adherence for TB medications?

A. Directly observed therapy (DOT)

B. Incentives

C. Combination products

D. Hospitalization

Answer: A, B, and C

TB treatment requires multiple medications for a prolonged period, so poor adherence can result in treatment failure, relapse, and development of drug resistance. Several strategies can make completion of treatment easier.

DOT involves observing the patient taking medication and can provide support and monitoring. Incentives and enablers can help overcome financial, transportation, or social barriers. Combination products reduce the number of individual pills and simplify the regimen.

Why Not Other Options?

D. Hospitalization ❌

Hospitalization may be necessary for severe TB, serious complications, or certain infection-control situations, but routinely admitting patients to hospital is not a standard strategy simply to improve medication adherence.

Key exam point

💡 TB adherence strategies → DOT + incentives/support + simplified drug regimens.

💡 Good adherence helps prevent treatment failure and drug resistance.

Remember

✓ DOT → supports correct medication use

✓ Combination therapy → reduces pill burden

✓ Long treatment → adherence is critical

22. Select the INCORRECT statement regarding respiratory infections, seasonal flu, and the common cold:

- A. The cold symptoms are usually milder than flu.
- B. Rhinovirus is the most common causative agent of the common cold.
- C. Flu vaccination is recommended annually for 6 months of age and older.
- D. There are no antiviral drugs available for the treatment of flu.

Answer: D. There are no antiviral drugs available for the treatment of flu.

The statement in D is incorrect because antiviral drugs are available for influenza. Examples include oseltamivir, zanamivir, and baloxavir. Antivirals are particularly useful when started early and are important for patients at increased risk of influenza complications.

The common cold is usually a milder upper respiratory infection, while influenza tends to cause more prominent systemic symptoms such as fever, myalgia, headache, and fatigue.

Why Not Other Options?

A. The cold symptoms are usually milder than flu. ❌

This is true. Colds commonly cause runny nose, sneezing, sore throat, and mild cough, while influenza more often causes abrupt fever, body aches, headache, and marked fatigue.

B. Rhinovirus is the most common causative agent of the common cold. ❌

This is true. Rhinoviruses are the most common cause, although many other respiratory viruses can also produce cold symptoms.

C. Flu vaccination is recommended annually for 6 months of age and older. ❌

This is true. Annual influenza vaccination is routinely recommended beginning at 6 months of age, with specific recommendations depending on age and clinical circumstances.

Key exam point

✦ Influenza → antiviral drugs are available.

✦ Rhinovirus → most common cause of the common cold.

✦ Flu → generally more systemic and severe than a common cold.

23. Which is NOT a typical clinical presentation of pulmonary tuberculosis?

- A. Pharyngitis
- B. Night sweating
- C. Haemoptysis (bloody sputum)
- D. Productive cough

Answer: A. Pharyngitis

Pulmonary TB usually develops gradually and commonly causes persistent cough, sputum production, fever, night sweats, weight loss, weakness, and sometimes hemoptysis. The symptoms can persist for weeks or months rather than appearing suddenly.

Hemoptysis is particularly associated with cavitory pulmonary disease, where inflammation and tissue destruction can damage pulmonary blood vessels.

Why Not Other Options?

B. Night sweating ❌

Night sweats are a classic constitutional symptom of TB and may occur with fever and weight loss.

C. Haemoptysis (bloody sputum) ✖

Hemoptysis can occur in pulmonary TB, especially when cavitory lesions damage surrounding blood vessels.

D. Productive cough ✖

A persistent cough, often with sputum, is one of the major clinical manifestations of pulmonary TB.

Key exam point

💡 Pulmonary TB → chronic cough + fever/night sweats + weight loss ± hemoptysis.

💡 Symptoms often develop gradually over weeks or months.

Remember

✓ Night sweats → classic TB clue

✓ Hemoptysis → possible with cavitory TB

✓ Pharyngitis → not typical pulmonary TB

24. A 54-year-old man with previous pulmonary TB has productive cough and hemoptysis. Imaging shows a fungal ball in a pre-existing lung cavity. What is the most likely causative agent?

- A. *Blastomyces dermatitidis*
- B. *Histoplasma capsulatum*
- C. *Coccidioides immitis*
- D. *Aspergillus fumigatus*

Answer: D. *Aspergillus fumigatus*

This presentation is classic for an aspergilloma, also called an *Aspergillus* fungal ball. *Aspergillus fumigatus* can colonize a pre-existing cavity in the lung without necessarily invading the surrounding tissue.

A previous pulmonary TB infection is an important risk factor because TB can leave residual cavities. *Aspergillus* hyphae, mucus, cellular debris, and other material can accumulate within these cavities and form a fungal ball.

A major clinical clue is hemoptysis, which occurs because the fungal mass and associated inflammation can damage blood vessels in the cavity wall.

Why Not Other Options?

A. *Blastomyces dermatitidis* ✗

Causes blastomycosis, which can involve the lungs and skin. It does not classically form a fungal ball inside an old pulmonary cavity.

B. *Histoplasma capsulatum* ✗

Can cause chronic pulmonary disease and cavitary lesions, particularly in older adults with underlying lung disease, but it is not the classic cause of a fungal ball.

C. *Coccidioides immitis* ✗

Can produce pulmonary nodules and cavities, but a fungal ball within a pre-existing cavity is much more characteristic of *Aspergillus*.

Key exam point

✦ Pre-existing lung cavity + fungal ball + hemoptysis → *Aspergillus aspergilloma*.

✦ Previous pulmonary TB is a classic risk factor.

Remember

✓ Old TB cavity → *Aspergillus* can colonize it

✓ Fungal ball → *Aspergilloma*

✓ *Aspergilloma* → may cause hemoptysis

25. Which bacteria is found in high concentration in the placenta of farm animals?

- A. *Streptococcus pyogenes*
- B. *Haemophilus influenzae*
- C. *Corynebacterium diphtheriae*
- D. *Coxiella burnetii*

Answer: D. *Coxiella burnetii*

Coxiella burnetii is the causative agent of Q fever, an important zoonotic disease associated with cattle, sheep, and goats. Infected animals can shed very large numbers of organisms in placenta, birth fluids, urine, feces, and milk, with particularly high concentrations in placental tissue.

Humans usually acquire infection by inhaling contaminated aerosols or dust rather than through direct animal-to-human transmission. Acute Q fever can cause a nonspecific febrile illness and may present with pneumonia or hepatitis. Chronic infection is classically associated with endocarditis.

Why Not Other Options?

A. *Streptococcus pyogenes* ❌

Group A *Streptococcus* is primarily a human pathogen, commonly causing pharyngitis, skin infections, and invasive disease. It is not the classic zoonotic organism associated with farm-animal placenta.

B. *Haemophilus influenzae* ❌

This is primarily a human respiratory pathogen, associated with infections such as otitis media, sinusitis, pneumonia, and meningitis. It is not characteristically associated with farm-animal placenta.

C. *Corynebacterium diphtheriae* ❌

Causes diphtheria, particularly respiratory disease in humans, and is not the characteristic pathogen associated with placenta of farm animals.

Key exam point

💡 *Coxiella burnetii* → Q fever → cattle, sheep, and goats.

💡 Very high concentrations can occur in placenta and birth products.

Remember

- ✓ Farm animals + placenta + zoonotic disease → *Coxiella burnetii*
- ✓ *Coxiella burnetii* → Q fever
- ✓ Inhalation of contaminated aerosols → major route of human infection

Exam:-Microbiology

Year:-2016 E.C

Dr. Doc 

1. A 24-year-old woman comes to the clinic with complaints of nasal discharge which is clear and watery and nasal obstruction. Physical examination reveals inflammatory edema of the nasal mucosa, and an increase in mucous secretions. No pain on palpation of the sinuses is noted. Her temperature is 37°C. Which of the following is the most common cause of this patient's condition?

- A. Rhinovirus
- B. Coxsackie B virus
- C. Echovirus
- D. All of the above

Answer: A. Rhinovirus

The patient's symptoms are typical of the common cold (acute viral rhinitis). Rhinovirus is the most common cause of the common cold. It primarily infects the upper respiratory tract and produces watery rhinorrhea, nasal congestion, sneezing, sore throat, and cough. Fever is usually absent or mild in adults.

The lack of sinus tenderness, high fever, or severe facial pain also makes acute bacterial sinusitis less likely.

▼ The patient's symptoms—clear watery nasal discharge, nasal obstruction, and a normal temperature (37°C)—are classic manifestations of rhinitis, the primary component of the common cold. While several viruses can cause this syndrome, rhinoviruses are explicitly identified as the most frequent infectious cause worldwide.

Why Not Other Options?

B. Coxsackie B virus ❌

Coxsackie B viruses are enteroviruses. They can occasionally cause respiratory symptoms, but they are more classically associated with myocarditis, pericarditis, and pleurodynia.

C. Echovirus ❌

Echoviruses can cause respiratory infections, especially in children, but are more commonly associated with fever, rash, aseptic meningitis, and other enteroviral illnesses. They are not the most common cause of the common cold.

D. All of the above ❌

Although Coxsackie B virus and echovirus can sometimes cause respiratory symptoms, rhinovirus is the most common cause of this clinical presentation.

Key exam point

✦ Common cold → Rhinovirus is the most common cause.

✦ Typical features → watery nasal discharge + nasal obstruction + sneezing ± sore throat.

Remember

✓ Rhinovirus → common cold

✓ Adults → fever usually absent or mild

✓ Watery rhinorrhea → classic upper respiratory viral infection

2. Antigenic variations in the surface glycoproteins hemagglutinin (HA) and neuraminidase (NA) are used to divide influenza viruses into:

A. Genera

B. Subfamilies

C. Subtypes

D. Types A, B, & C

Answer: C. Subtypes

▼ Influenza viruses have two major surface glycoproteins: hemagglutinin (HA) and neuraminidase (NA). Different combinations of HA and NA are used to identify influenza subtypes, particularly in influenza A.

For example, H1N1 means the virus has HA type 1 and NA type 1, while H3N2 has HA type 3 and NA type 2.

▼ These surface proteins are important because they are major targets of the host immune response and can undergo antigenic changes.

▼ Influenza viruses are divided into types (A, B, and C) based on their internal nucleoprotein and matrix proteins.

▼ The surface glycoproteins Hemagglutinin (HA) and Neuraminidase (NA) are used specifically to categorize these viruses into subtypes (e.g., H1N1, H3N2).

Why Not Other Options?

A. Genera ✗

Genus is a much broader taxonomic classification. HA and NA variations are not used to define influenza genera.

B. Subfamilies ✗

Subfamily is also a broad taxonomic level and does not describe the H/N classification.

D. Types A, B, & C ✗

Influenza viruses are classified into types A, B, C, and D based mainly on their internal viral proteins, rather than their HA and NA combinations. HA/NA combinations are used to identify subtypes, especially for influenza A.

Key exam point

✦ HA + NA → influenza subtype classification.

✦ Examples: H1N1 and H3N2.

Remember

✓ H = Hemagglutinin

✓ N = Neuraminidase

✓ H1N1 = subtype

3. The following information is indicated in the standard nomenclature system for human isolates of influenza virus EXCEPT:

A. Geographic origin

B. Type

C. Host of origin

D. Year of isolation

Answer: C. Host of origin

Influenza virus nomenclature provides information that helps identify the virus isolate. It includes the virus type, geographic location where it was isolated, strain/serial number, and year of isolation.

For human influenza isolates, the host is understood to be human and therefore is generally not included in the standard designation.

For example:

A/California/04/2009

This represents type A / geographic origin / isolate number / year.

Why Not Other Options?

A. Geographic origin ❌

The location where the virus was isolated is included in the nomenclature.

B. Type ❌

The influenza type, such as A or B, is included.

D. Year of isolation ❌

The year in which the virus was isolated is included.

Key exam point

💡 Influenza nomenclature includes type + geographic origin + isolate number + year.

Remember

✓ A/California/04/2009

✓ Type → Location → Strain number → Year

4. Minor antigenic changes in influenza viruses are:

- A. Result in a new subtype
- B. Due to genetic reassortment
- C. The cause of pandemics
- D. Result in amino acid changes in the protein

Answer: D. Result in amino acid changes in the protein

Minor antigenic changes are called antigenic drift. They occur because influenza viruses, especially influenza A and B, accumulate small mutations (point mutations) in genes encoding surface proteins such as HA and NA.

These mutations can change amino acids in the viral proteins, allowing the virus to partially escape pre-existing immunity. Antigenic drift is an important reason why influenza continues to circulate and why seasonal influenza vaccines are updated periodically.

Why Not Other Options?

A. Result in a new subtype ✗

Antigenic drift produces small changes within an existing subtype. A major change that creates a new subtype is associated with antigenic shift.

B. Due to genetic reassortment ✗

Genetic reassortment is the mechanism of antigenic shift, not antigenic drift.

C. The cause of pandemics ✗

Major influenza pandemics are associated with antigenic shift, which can produce a novel influenza A subtype to which the population has limited immunity.

Key exam point

✧ Antigenic drift → small mutations → gradual antigenic change → seasonal epidemics.

▲ Minor antigenic changes, known as antigenic drift, result from the accumulation of point mutations in the HA or NA genes. These mutations lead to amino acid changes in the surface proteins, which can allow the virus to escape recognition by the host's immune system. In contrast, the formation of new subtypes and pandemics are associated with major changes (antigenic shift).

Remember

✓ Drift = small, gradual changes

✓ Point mutation → drift

✓ Seasonal influenza → drift

5. Which factor does NOT contribute to antigenic shift in influenza viruses?

- A. Accumulation of point mutations in the gene
- B. Segmented nature of the viral genome
- C. Coinfection of a host cell by two different viruses of a given type
- D. Genetic reassortment

Answer: A. Accumulation of point mutations in the gene

Antigenic shift is a major and abrupt change in influenza virus antigens. It occurs mainly through genetic reassortment, where genome segments from two different influenza viruses mix within a coinfecting host cell.

Influenza A is particularly capable of antigenic shift because its genome consists of 8 separate RNA segments. If two influenza viruses infect the same cell, their genome segments can be packaged in new combinations, potentially producing a virus with a novel HA or NA.

Point mutations, however, are responsible for antigenic drift, not shift.

Why Not Other Options?

B. Segmented nature of the viral genome ❌

The segmented genome allows influenza viruses to exchange entire RNA segments, making reassortment possible.

C. Coinfection of a host cell by two different viruses of a given type ❌

Coinfection allows the genome segments of the two viruses to mix during replication and packaging.

D. Genetic reassortment ❌

Reassortment is the central mechanism of antigenic shift. It can produce a virus with a new combination of surface antigens.

Key exam point

💡 Antigenic shift → genetic reassortment of segmented influenza genomes.

💡 It can produce a new influenza A subtype and potentially cause a pandemic.

Remember

✓ Point mutation → Antigenic drift

✓ Genome reassortment → Antigenic shift

✓ Segmented genome + coinfection → shift

6. Bacterial etiologic agents of pharyngitis or tonsillitis include:

A. *Streptococcus pyogenes*

B. *Corynebacterium diphtheriae*

C. *Neisseria gonorrhoeae*

D. All of the above

Answer: D. All of the above

Pharyngitis and tonsillitis can be caused by several bacteria. *Streptococcus pyogenes* (Group A *Streptococcus*) is the most important and common bacterial cause of acute pharyngitis, especially in children. *Corynebacterium diphtheriae* causes pharyngeal diphtheria with a characteristic gray pseudomembrane, while *Neisseria gonorrhoeae* can cause gonococcal pharyngitis, particularly following orogenital exposure.

Why Not Other Options?

A. *Streptococcus pyogenes* ✗

It is a major cause of bacterial pharyngitis and can produce complications such as rheumatic fever and post-streptococcal glomerulonephritis.

B. *Corynebacterium diphtheriae* ✖

It can cause pharyngeal diphtheria, characterized by a firmly adherent gray-white pseudomembrane and cervical lymphadenopathy.

C. *Neisseria gonorrhoeae* ✖

It can infect the pharynx and cause gonococcal pharyngitis, although this is less common than streptococcal pharyngitis.

Key exam point

💡 Bacterial pharyngitis → *S. pyogenes* is the classic/common cause.

💡 *C. diphtheriae* and *N. gonorrhoeae* can also cause pharyngeal disease.

Remember

✓ *S. pyogenes* → pharyngitis

✓ *C. diphtheriae* → pseudomembrane

✓ *N. gonorrhoeae* → sexually associated pharyngitis

7. Which virulence factor of *S. pyogenes* is responsible for the characteristic β -hemolysis on blood agar?

- A. Streptolysin O
- B. Streptolysin S
- C. Streptokinase
- D. Deoxyribonuclease

Answer: B. Streptolysin S

Streptococcus pyogenes produces two important hemolysins: streptolysin O (SLO) and streptolysin S (SLS). Streptolysin S is responsible for the characteristic clear β -hemolysis seen around colonies on blood agar under routine aerobic conditions.

Streptolysin O is oxygen-labile and is important clinically because antibodies against it produce the ASO (antistreptolysin O) titer, which can help demonstrate a recent streptococcal infection.

Why Not Other Options?

A. Streptolysin O ❌

It is also a hemolysin, but it is oxygen-labile and is not primarily responsible for the readily visible β -hemolysis on the surface of routine blood agar.

C. Streptokinase ❌

Streptokinase promotes fibrinolysis by activating plasminogen to plasmin. It does not cause hemolysis.

D. Deoxyribonuclease ❌

DNase breaks down extracellular DNA and contributes to spread through tissues, but it does not produce the characteristic β -hemolysis.

Key exam point

✦ S. pyogenes β -hemolysis $\rightarrow$ Streptolysin S.

✦ ASO antibody $\rightarrow$ Streptolysin O exposure.

Remember

✓ SLS $\rightarrow$ β -hemolysis

✓ SLO $\rightarrow$ ASO titer

8. Initial attachment of *Neisseria gonorrhoeae* to the host cell is mediated by:

A. Opa protein

B. Pili

C. Por protein

D. Lipooligosaccharide

Answer: B. Pili

The type IV pili of *N. gonorrhoeae* are important for the initial attachment of the organism to mucosal epithelial cells. They allow the bacteria to adhere to the host surface and are also involved in twitching motility and antigenic variation.

After initial attachment, other surface proteins, particularly Opa proteins, contribute to closer interaction with and invasion of host cells.

Why Not Other Options?

A. Opa protein ✗

Opa proteins are important for tight adherence and interaction with host cells, particularly after the initial pilus-mediated attachment.

C. Por protein ✗

Porins form channels in the outer membrane and contribute to nutrient transport and immune evasion, rather than serving as the primary initial attachment structure.

D. Lipooligosaccharide ✗

LOS contributes strongly to inflammation and tissue damage and has endotoxin-like activity, but it is not the primary mediator of initial attachment.

Key exam point

💡 *N. gonorrhoeae* initial attachment → type IV pili.

Remember

✓ Pili → initial attachment

✓ Opa → tighter adhesion/invasion

✓ LOS → inflammation

9. The limitation of direct antigen tests for rapidly detecting *S. pyogenes* in throat swabs is:

- A. Lack of sensitivity
- B. False-positive result
- C. Lack of specificity
- D. All of the above

Answer: A. Lack of sensitivity

Rapid antigen detection tests (RADTs) for Group A Streptococcus are useful because they provide results quickly. Their major limitation is imperfect sensitivity, meaning some patients with true *S. pyogenes* infection may have a negative rapid test.

Their specificity is generally high, so a positive result is usually reliable. Because of the sensitivity limitation, a negative test may require confirmatory testing in certain patient groups, particularly children, depending on clinical guidelines.

Why Not Other Options?

B. False-positive result ✗

False positives can occur, but they are not the major limitation. The high specificity of these tests makes positive results generally reliable.

C. Lack of specificity ✗

The tests are generally highly specific for Group A Streptococcus.

D. All of the above ✗

The major limitation is sensitivity, not a general lack of specificity.

Key exam point

💡 Rapid strep antigen test → fast + highly specific, but sensitivity can be limited.

Remember

✓ Positive RADT → usually reliable

✓ Negative RADT → sensitivity limitation must be considered

10. Acute diseases of the middle airway are usually caused by viral agents EXCEPT:

A. Bronchitis

B. Laryngitis

C. Epiglottitis

D. Laryngotracheitis

Answer: C. Epiglottitis

Bronchitis, laryngitis, and laryngotracheitis (croup) are commonly caused by viruses. Epiglottitis, however, has traditionally been associated with bacterial infection, especially *Haemophilus influenzae* type b (Hib) in unvaccinated children. Other bacteria, including *Streptococcus* and *Staphylococcus*, can also cause it.

Hib vaccination has dramatically reduced childhood Hib epiglottitis.

Why Not Other Options?

A. Bronchitis ❌

Acute bronchitis is predominantly viral, with viruses such as influenza, parainfluenza, RSV, and rhinovirus being common causes.

B. Laryngitis ❌

Acute laryngitis is usually caused by a viral upper respiratory infection and commonly produces hoarseness.

D. Laryngotracheitis ❌

Viral laryngotracheitis is commonly called croup and is classically associated with parainfluenza viruses.

Key exam point

💡 Croup/laryngotracheitis → Parainfluenza virus.

💡 Epiglottitis → think bacterial, especially Hib in an unvaccinated child.

Remember

✓ Croup → barking cough + stridor → Parainfluenza

✓ Epiglottitis → airway emergency

11. Growth of which bacterium requires supplementation of media with hemin and NAD?

- A. *Bordetella pertussis*
- B. *Haemophilus influenzae*
- C. *Streptococcus pneumoniae*
- D. *Chlamydophila pneumoniae*

Answer: B. *Haemophilus influenzae*

Haemophilus influenzae requires two important growth factors: X factor (hemin) and V factor (NAD, nicotinamide adenine dinucleotide).

This requirement is an important laboratory identification feature. *H. influenzae* can grow on chocolate agar, where red blood cells are lysed and these growth factors become available.

It also demonstrates the classic phenomenon of satellitism around *Staphylococcus aureus* on blood agar because *S. aureus* can provide V factor and other nutrients.

Why Not Other Options?

A. *Bordetella pertussis* ✗

Requires specialized media such as Bordet-Gengou or Regan-Lowe agar, rather than X and V factors.

C. *Streptococcus pneumoniae* ✗

Can grow on blood agar and characteristically produces α -hemolysis, optochin sensitivity, and bile solubility.

D. *Chlamydophila pneumoniae* ✗

It is an obligate intracellular bacterium and requires living host cells for growth rather than routine X/V factor-supplemented media.

Key exam point

💡 H. influenzae → X factor (hemin) + V factor (NAD).

Remember

✓ X = hemin

✓ V = NAD

✓ H. influenzae → chocolate agar

12. Haemophilus influenzae is subdivided into eight biotypes as determined by:

- A. Indole production
- B. Urease activity
- C. Ornithine decarboxylase activity
- D. All of the above

Answer: D. All of the above

Haemophilus influenzae can be divided into 8 biotypes (I–VIII) based on a combination of biochemical reactions. The three important tests are indole production, urease activity, and ornithine decarboxylase activity. The pattern of positive or negative reactions helps laboratories distinguish the different biotypes.

It is important to distinguish biotyping from serotyping. Biotyping uses biochemical characteristics, whereas serotyping identifies strains based on their capsular antigens. The clinically important encapsulated strain H. influenzae type b (Hib) is a serotype, not a biotype.

Why Not Other Options?

A. Indole production ❌

Indole production is one of the biochemical tests used for biotyping, but it does not determine the biotype by itself.

B. Urease activity ❌

Urease activity is also used in the biotyping scheme, but it must be considered together with the other biochemical reactions.

C. Ornithine decarboxylase activity ❌

This is another component of the biotyping system, but it is not sufficient alone to classify all eight biotypes.

Key exam point

💡 H. influenzae biotypes I–VIII are determined using three biochemical reactions: indole, urease, and ornithine decarboxylase.

💡 Biotype = biochemical characteristics; serotype = capsule.

Remember

✓ Indole + Urease + Ornithine decarboxylase → Biotyping

✓ 8 biotypes → I–VIII

✓ Hib → serotype b, based on capsule

13. The risk of epiglottitis caused by *Haemophilus influenzae* is significantly greater in:

- A. Patients with no anti-PRP antibodies
- B. Those with depletion of complement
- C. Those who have undergone splenectomy
- D. All of the above

Answer: D. All of the above

Haemophilus influenzae type b (Hib) is an encapsulated bacterium whose capsule contains polyribosylribitol phosphate (PRP). Anti-PRP antibodies are important for protection because they promote opsonization and clearance of Hib.

Patients who lack effective anti-PRP antibodies are therefore at increased risk. Complement is also important for bacterial killing, while the spleen plays a major role in clearing encapsulated organisms. Consequently, asplenic or splenectomized patients are particularly vulnerable to invasive Hib infection.

Why Not Other Options?

A. Patients with no anti-PRP antibodies ❌

This is a genuine risk factor because anti-PRP antibodies help the immune system recognize and eliminate encapsulated Hib. However, it is not the only risk factor listed.

B. Those with depletion of complement ❌

Complement contributes to opsonization and immune defense against encapsulated bacteria. Complement deficiency can therefore increase susceptibility to invasive bacterial infections.

C. Those who have undergone splenectomy ❌

The spleen is particularly important for removing encapsulated organisms from the bloodstream. Splenectomy therefore increases the risk of invasive infections.

Key exam point

💡 Hib is encapsulated → effective antibody, complement, and splenic function are important for protection.

Remember

✓ Hib capsule → PRP

✓ Asplenia → increased risk from encapsulated bacteria

✓ Anti-PRP antibodies → protection against Hib

14. Which virulence factor facilitates subsequent migration of *Streptococcus pneumoniae* to the lower respiratory tract?

- A. Capsule
- B. Secretory IgA protease
- C. Teichoic acid
- D. All of the above

Answer: D. All of the above

Several pneumococcal factors contribute to successful colonization and progression from the nasopharynx into the lower respiratory tract. The capsule prevents phagocytosis and is the major virulence factor. IgA protease helps the organism survive on respiratory mucosal surfaces by degrading secretory IgA. Teichoic acids contribute to adherence and inflammatory interactions with host tissues.

Together, these factors help *S. pneumoniae* colonize the respiratory tract and potentially progress to pneumonia.

Why Not Other Options?

A. Capsule ✗

The capsule is extremely important because it protects pneumococci from phagocytosis, but it is not the only factor involved.

B. Secretory IgA protease ✗

IgA protease helps the organism evade mucosal immunity, but it is only one component of pneumococcal virulence.

C. Teichoic acid ✗

Teichoic acids participate in adhesion and inflammatory responses, but they do not act alone to explain respiratory tract progression.

Key exam point

✦✦ Pneumococcal capsule is the major virulence factor.

✦✦ IgA protease helps pneumococci overcome mucosal immune defenses.

Remember

✓ Capsule → antiphagocytic

✓ IgA protease → mucosal immune evasion

✓ Teichoic acid → adhesion/inflammation

15. Tissue destruction by *Streptococcus pneumoniae* is mediated by:

A. Teichoic acid

B. Peptidoglycan fragments

C. Pneumolysin

D. All of the above

Answer: D. All of the above

Pneumococcal tissue injury results from several bacterial components. Pneumolysin is a major toxin that forms pores in host cell membranes and damages epithelial cells and immune cells. Peptidoglycan fragments and teichoic acids can stimulate a strong inflammatory response, contributing to tissue injury.

Therefore, pneumococcal disease is produced not only by direct bacterial effects but also by the host inflammatory response triggered by bacterial cell-wall components.

Why Not Other Options?

A. Teichoic acid ❌

Teichoic acids can stimulate inflammation and contribute to host tissue injury, but they are not the only mediator.

B. Peptidoglycan fragments ❌

These cell-wall components activate inflammatory pathways and contribute to tissue damage, but they act together with other bacterial factors.

C. Pneumolysin ❌

Pneumolysin is a major cytotoxic factor, but tissue destruction also involves inflammatory effects of cell-wall components.

Key exam point

💡 Pneumococcal tissue injury → pneumolysin + inflammatory effects of cell-wall components.

Remember

✓ Pneumolysin → pore-forming toxin

✓ Teichoic acid/peptidoglycan → inflammation

16. Which virulence factor of *S. aureus* functions as a superantigen?

A. Toxic shock syndrome toxin-1 (TSST-1)

B. Panton-Valentine (P-V) leukocidin

C. Alpha toxin

D. Sphingomyelinase C

Answer: A. Toxic shock syndrome toxin-1 (TSST-1)

TSST-1 is a classic superantigen produced by some strains of *Staphylococcus aureus*. Unlike conventional antigens, superantigens cause widespread, nonspecific activation of T cells by cross-linking MHC class II molecules on antigen-presenting cells with T-cell receptors.

This results in massive cytokine release and can produce fever, hypotension, rash, multiorgan dysfunction, and toxic shock syndrome.

Why Not Other Options?

B. Panton-Valentine leukocidin ✗

PVL is a toxin that damages neutrophils and other leukocytes and is associated with severe skin infections and some necrotizing pneumonia. It is not a superantigen.

C. Alpha toxin ✗

Alpha toxin is a pore-forming cytotoxin that damages host cell membranes. It does not cause superantigen-mediated T-cell activation.

D. Sphingomyelinase C ❌

This enzyme damages cell membranes by hydrolyzing sphingomyelin. It is not a superantigen.

Key exam point

💡 S. aureus superantigen → TSST-1.

Remember

✓ TSST-1 → toxic shock

✓ PVL → leukocyte destruction

✓ Alpha toxin → membrane damage

17. The diffuse vascular damage associated with *Neisseria meningitidis* infection is largely attributed to:

A. Meningococcal pili

B. Capsule

C. Lipooligosaccharide (LOS)

D. All of the above

Answer: C. Lipooligosaccharide (LOS)

The major cause of vascular injury and shock in meningococcal sepsis is the endotoxin-like activity of lipooligosaccharide (LOS) in the outer membrane of *N. meningitidis*.

When meningococci multiply and release LOS, it triggers a powerful inflammatory response with release of cytokines and activation of coagulation pathways. Severe cases can develop septic shock, disseminated intravascular coagulation (DIC), petechiae/purpura, and widespread vascular injury.

Why Not Other Options?

A. Meningococcal pili ❌

Pili primarily help *N. meningitidis* attach to and colonize nasopharyngeal epithelial cells. They are not the main cause of vascular damage.

B. Capsule ❌

The capsule protects the organism from phagocytosis and complement-mediated clearance, helping it survive in the bloodstream, but it is not responsible for the major inflammatory vascular injury.

D. All of the above ❌

Although pili and capsule contribute to meningococcal virulence, LOS is the major mediator of endotoxin-induced vascular damage.

Key exam point

💡 *N. meningitidis* LOS → massive inflammation → vascular injury, shock, and DIC.

Remember

✓ Pili → attachment

✓ Capsule → antiphagocytic protection

✓ LOS → endotoxin + shock/DIC

18. A 2-year-old unvaccinated child has dysphonia, grayish exudate over the tonsils, larynx and pharynx, and massive tender cervical lymphadenitis. What is the most likely cause?

A. *Corynebacterium diphtheriae*

B. *Streptococcus pneumoniae*

C. *Bordetella pertussis*

D. *Haemophilus influenzae*

Answer: A. *Corynebacterium diphtheriae*

This is a classic presentation of respiratory diphtheria. The combination of an unvaccinated child, sore throat, dysphonia, gray adherent pseudomembrane, and cervical lymphadenopathy is highly characteristic.

C. *diphtheriae* produces diphtheria toxin, which inhibits protein synthesis by inactivating elongation factor-2 (EF-2). The toxin causes local tissue necrosis and pseudomembrane formation and can produce systemic complications such as myocarditis and peripheral neuropathy.

Why Not Other Options?

B. *Streptococcus pneumoniae* ✗

It commonly causes pneumonia, otitis media, sinusitis, and meningitis. It does not characteristically produce the thick gray pseudomembrane of diphtheria.

C. *Bordetella pertussis* ❌

Causes pertussis (whooping cough), characterized by prolonged paroxysmal coughing and inspiratory whoop rather than a gray pharyngeal pseudomembrane.

D. *Haemophilus influenzae* ❌

Hib can cause epiglottitis, meningitis, and other invasive infections, but the classic gray pseudomembrane and cervical lymphadenopathy point toward diphtheria.

Key exam point

💡 Unvaccinated child + gray pseudomembrane + cervical lymphadenopathy → diphtheria.

💡 Diphtheria toxin inhibits EF-2 → blocks protein synthesis.

Remember

✓ Gray pseudomembrane → *C. diphtheriae*

✓ “Bull neck” → severe cervical swelling in diphtheria

✓ Diphtheria toxin → EF-2 inhibition

19. Which of the following bacteria is an obligate aerobe, a highly contagious pathogen that may cause serious respiratory infections in young children?

- A. *Haemophilus ducreyi*
- B. *Francisella tularensis*
- C. *Haemophilus parainfluenzae*
- D. *Bordetella pertussis*

Answer: D. *Bordetella pertussis*

Bordetella pertussis is a small, Gram-negative coccobacillus and a strict/obligate aerobe. It causes pertussis (whooping cough), a highly contagious respiratory infection that particularly affects infants and young children.

The disease typically begins with a catarrhal stage resembling a common cold, followed by a paroxysmal stage with repeated severe coughing episodes, often followed by an inspiratory “whoop.” Vomiting after coughing can also occur. Its major virulence factors include pertussis toxin, adenylate cyclase toxin, tracheal cytotoxin, and filamentous hemagglutinin.

Why Not Other Options?

A. *Haemophilus ducreyi* ❌

Causes chancroid, a sexually transmitted genital ulcer disease, rather than whooping cough or a typical respiratory infection.

B. *Francisella tularensis* ❌

Causes tularemia, a zoonotic infection that can occasionally produce pneumonia, but it is not the classic highly contagious childhood respiratory pathogen causing whooping cough.

C. *Haemophilus parainfluenzae* ❌

It is part of the normal respiratory flora and can occasionally cause opportunistic infections, but it is not the causative organism of pertussis.

Key exam point

✦ Bordetella pertussis → highly contagious respiratory infection → whooping cough.

✦ It is a Gram-negative, small coccobacillus and obligate aerobe.

Remember

✓ B. pertussis → pertussis

✓ Paroxysmal cough + whoop → pertussis

✓ Infants and young children → important risk group

20. A 69-year-old woman has fever, body aches, mild cough, and lobar pneumonia. Round diplococci in chains are isolated. What is the likely pathogen?

A. Streptococcus pneumoniae

B. Neisseria gonorrhoeae

C. Haemophilus influenzae

D. Bordetella pertussis

Answer: A. Streptococcus pneumoniae

The description strongly points to Streptococcus pneumoniae. It is a Gram-positive, lancet-shaped, encapsulated diplococcus, classically arranged in pairs and short chains.

It is the classic leading bacterial cause of community-acquired pneumonia and commonly produces lobar pneumonia. Typical symptoms include fever, productive cough, pleuritic chest pain, and sometimes rusty-colored sputum.

Its polysaccharide capsule is its most important virulence factor because it prevents phagocytosis. Pneumococci can also cause otitis media, sinusitis, and meningitis.

Why Not Other Options?

B. *Neisseria gonorrhoeae* ❌

This is a Gram-negative diplococcus primarily associated with gonorrhea, including urethritis and cervicitis. It does not typically cause classic lobar pneumonia.

C. *Haemophilus influenzae* ❌

It can cause pneumonia, especially in older adults and people with chronic lung disease, but it is a small Gram-negative coccobacillus, not a Gram-positive diplococcus.

D. *Bordetella pertussis* ❌

Causes pertussis with prolonged paroxysmal coughing, not typical lobar pneumonia with pneumococcal morphology.

Key exam point

💡 Lobar pneumonia + Gram-positive lancet-shaped diplococci → *S. pneumoniae*.

Remember

✓ *S. pneumoniae* → Gram-positive diplococcus

✓ Encapsulated → antiphagocytic

✓ Classic bacterial CAP → *S. pneumoniae*

21. Which human fungal species, also present in the lungs of animals, causes pneumonia in immunocompromised patients?

A. *Aspergillus niger*

B. *Pneumocystis jirovecii*

C. *Penicillium marneffei*

D. *Fusarium keratitis*

Answer: B. *Pneumocystis jirovecii*

Pneumocystis jirovecii is an opportunistic fungal organism that causes *Pneumocystis pneumonia* (PCP), especially in severely immunocompromised patients, classically those with advanced HIV/AIDS and low CD4 counts.

It is unusual because *Pneumocystis* was historically considered a protozoan but is now classified as a fungus. The infection typically produces progressive dry cough, fever, and shortness of breath, with diffuse bilateral interstitial or ground-glass pulmonary abnormalities.

Why Not Other Options?

A. *Aspergillus niger* ✗

Aspergillus species can cause pulmonary disease in immunocompromised patients, but *A. niger* is not the classic organism responsible for PCP.

C. *Penicillium marneffei* ❌

Now called *Talaromyces marneffei*, it causes talaromycosis, particularly in immunocompromised individuals in Southeast Asia. It is not the classic cause of this pneumonia.

D. *Fusarium* keratitis ❌

Fusarium can cause keratitis and invasive infections in severely immunocompromised patients, but it is not the classic cause of opportunistic pneumonia described here.

Key exam point

💡 Immunocompromised patient + diffuse pneumonia → think *Pneumocystis jirovecii*.

Remember

✓ *P. jirovecii* → PCP

✓ Major risk → severe immunosuppression

✓ Usually dry cough + progressive dyspnea

22. “Whooping cough” infection of the respiratory tract in children is caused by which Gram-negative pathogen?

A. *Bordetella pertussis*

B. *Mycoplasma pneumoniae*

C. *Mycobacterium tuberculosis*

D. *Salmonella typhi*

Answer: A. *Bordetella pertussis*

Pertussis, or whooping cough, is caused by *Bordetella pertussis*. It is a Gram-negative coccobacillus that primarily infects the respiratory tract.

The illness classically progresses through three stages: catarrhal, paroxysmal, and convalescent. The paroxysmal stage is characterized by repeated bouts of coughing followed by a characteristic inspiratory whoop, although the whoop may be absent in young infants.

Why Not Other Options?

B. *Mycoplasma pneumoniae* ❌

Causes atypical pneumonia, particularly in school-aged children and young adults. It lacks a cell wall and is not the cause of whooping cough.

C. *Mycobacterium tuberculosis* ❌

Causes tuberculosis, typically characterized by a chronic cough, fever, night sweats, weight loss, and sometimes hemoptysis.

D. *Salmonella typhi* ❌

Causes typhoid fever, primarily an enteric/systemic infection rather than whooping cough.

Key exam point

💡 Whooping cough → *Bordetella pertussis*.

Remember

✓ B. pertussis → Gram-negative coccobacillus

✓ Paroxysmal cough → inspiratory whoop

✓ Highly contagious respiratory infection

23. Among the following Gram-negative bacteria, which of the following has been identified as the primary causative agent of hospital-acquired pneumonia in patients within healthcare facilities?

A. Escherichia coli

B. Bordetella pertussis

C. Pseudomonas aeruginosa

D. Yersinia pestis

Answer: C. Pseudomonas aeruginosa

Pseudomonas aeruginosa is an important cause of hospital-acquired pneumonia (HAP) and particularly ventilator-associated pneumonia (VAP). It is an aerobic, Gram-negative rod that can survive in moist hospital environments and is often associated with patients who have prolonged hospitalization, mechanical ventilation, or underlying lung disease.

It is also important because it can develop multiple antibiotic-resistance mechanisms, making treatment challenging.

Why Not Other Options?

A. *Escherichia coli* ❌

E. coli can cause hospital-acquired infections, including pneumonia, especially in critically ill patients, but it is not the classic organism associated with HAP in this question.

B. *Bordetella pertussis* ❌

Primarily causes community-acquired pertussis, not hospital-acquired pneumonia.

D. *Yersinia pestis* ❌

Causes plague, including pneumonic plague, but it is not a typical cause of hospital-acquired pneumonia.

Key exam point

💡 Hospital-acquired/ventilator-associated pneumonia → think *Pseudomonas aeruginosa*.

Remember

✓ *Pseudomonas* → Gram-negative rod

✓ HAP/VAP → important pathogen

✓ Often multidrug resistant

24. Which bacterium is a common cause of chronic lower respiratory tract infection?

- A. *Clostridium difficile*
- B. *Mycobacterium tuberculosis*
- C. *Moraxella catarrhalis*
- D. *Staphylococcus epidermidis*

Answer: B. *Mycobacterium tuberculosis*

Mycobacterium tuberculosis is a major cause of chronic lower respiratory tract disease. Pulmonary TB usually develops gradually and may cause persistent cough, sputum production, fever, night sweats, weight loss, and hemoptysis.

The organism is an acid-fast bacillus because its cell wall contains large amounts of mycolic acid. Chronic pulmonary infection can produce granulomas, caseous necrosis, and sometimes cavitory lesions, particularly in the upper lobes.

Why Not Other Options?

A. *Clostridium difficile* ✗

Primarily causes antibiotic-associated diarrhea and pseudomembranous colitis, not chronic respiratory infection.

C. *Moraxella catarrhalis* ✗

Can cause respiratory infections, particularly otitis media, sinusitis, and exacerbations of COPD, but it is not the classic cause of chronic pulmonary infection in this context.

D. *Staphylococcus epidermidis* ✗

It is mainly a skin flora organism and an important cause of infections involving indwelling medical devices and prosthetic material, rather than chronic pulmonary TB-like disease.

Key exam point

✦ Chronic pulmonary infection + night sweats/weight loss ± hemoptysis → think TB.

✦ *M. tuberculosis* → acid-fast bacillus with a mycolic-acid-rich cell wall.

Remember

✓ TB → chronic cough + night sweats + weight loss

✓ Mycolic acid → acid-fastness

✓ Pulmonary TB → may form cavities

25. TB disease is difficult to control and prevent mainly because of diagnostic challenges. Which technology is most amenable to a point-of-care testing program where treatment decisions can be made in the same visit?

- A. Smear microscopy
- B. GeneXpert MTB/RIF
- C. Culture
- D. Serological test

Answer: B. GeneXpert MTB/RIF

GeneXpert MTB/RIF is a rapid molecular test that can detect *Mycobacterium tuberculosis* and, importantly, identify rifampicin resistance. Results can be available much faster than culture, making it particularly useful when treatment decisions need to be made during the same clinical visit.

It uses PCR-based molecular detection rather than waiting for the organism to grow. This is especially valuable because *M. tuberculosis* grows slowly and conventional culture can take weeks.

Why Not Other Options?

A. Smear microscopy ❌

It is relatively rapid and inexpensive but has limited sensitivity, especially in patients with low bacterial loads. It also cannot reliably identify drug resistance.

C. Culture ❌

Culture is highly important and remains a reference method, but *M. tuberculosis* grows slowly, so it is not ideal for same-visit treatment decisions.

D. Serological test ❌

Serological tests are not reliable for diagnosing active TB because antibody responses are variable and they cannot adequately distinguish active disease from previous exposure.

Key exam point

💡 Rapid molecular TB diagnosis + rifampicin resistance → GeneXpert MTB/RIF.

Remember

✓ GeneXpert → rapid molecular test

✓ Detects MTB + rifampicin resistance

✓ Much faster than culture

26. Which of the following is NOT caused by a dimorphic fungus?

A. Histoplasmosis

B. Coccidioidomycosis

C. Blastomycosis

D. Aspergillosis

Answer: D. Aspergillosis

Histoplasma, Coccidioides, and Blastomyces are classic medically important dimorphic fungi. They can exist in different forms depending on environmental conditions, particularly temperature.

Aspergillus, in contrast, is a filamentous mold and is not classically classified as a thermally dimorphic fungus. It produces characteristic septate hyphae with acute-angle branching.

Why Not Other Options?

A. Histoplasmosis ✗

Caused by Histoplasma capsulatum, a classic dimorphic fungus associated with exposure to soil contaminated with bird or bat droppings.

B. Coccidioidomycosis ✗

Caused by *Coccidioides* species, which have a mold form in the environment and form spherules containing endospores in tissues.

C. Blastomycosis ❌

Caused by *Blastomyces dermatitidis*, a dimorphic fungus classically producing broad-based budding yeast in tissue.

Key exam point

💡 *Histoplasma* + *Coccidioides* + *Blastomyces* → classic dimorphic fungi.

💡 *Aspergillus* → mold, not a classic dimorphic fungus.

Remember

✓ *Histo* → intracellular yeast

✓ *Coccidioides* → spherules

✓ *Blastomyces* → broad-based budding

✓ *Aspergillus* → septate hyphae, acute-angle branching

27. The initial therapy of tuberculosis includes which drug regimen?

- A. Streptomycin and rifampin
- B. Isoniazid, streptomycin, and ethambutol
- C. Rifampin, isoniazid, and ciprofloxacin
- D. Isoniazid, rifampin, pyrazinamide, and ethambutol

Answer: D. Isoniazid, rifampin, pyrazinamide, and ethambutol

The classic initial treatment for drug-susceptible active pulmonary tuberculosis is the four-drug regimen:

Isoniazid + Rifampin + Pyrazinamide + Ethambutol (HRZE).

Using multiple drugs is important because *M. tuberculosis* can develop resistance when treatment relies on a single drug. The four drugs attack the organism through different mechanisms and help prevent the emergence of resistance.

Why Not Other Options?

A. Streptomycin and rifampin ❌

Only two drugs are listed and this is not the standard initial four-drug regimen for drug-susceptible TB.

B. Isoniazid, streptomycin, and ethambutol ❌

This lacks rifampin and pyrazinamide, which are important components of standard initial therapy.

C. Rifampin, isoniazid, and ciprofloxacin ❌

Ciprofloxacin is not part of the standard first-line four-drug initial regimen for uncomplicated drug-susceptible TB.

Key exam point

✦ Initial drug-susceptible TB regimen → HRZE.

Remember

✓ H → Isoniazid

✓ R → Rifampin

✓ Z → Pyrazinamide

✓ E → Ethambutol

28. Which bacterial pathogen is the most common cause of community-acquired pneumonia in adults?

A. *Staphylococcus aureus*

B. *Mycoplasma pneumoniae*

C. *Streptococcus pneumoniae*

D. *Chlamydophila pneumoniae*

Answer: C. *Streptococcus pneumoniae*

Streptococcus pneumoniae is the classic and most common bacterial cause of community-acquired pneumonia (CAP) in adults. It is an encapsulated, Gram-positive, lancet-shaped diplococcus.

Pneumococcal pneumonia often presents with fever, productive cough, pleuritic chest pain, and lobar consolidation. The capsule is its major virulence factor because it protects the organism from phagocytosis.

Why Not Other Options?

A. *Staphylococcus aureus* ✗

Can cause severe pneumonia, especially following influenza, but is not the classic most common bacterial cause of CAP.

B. *Mycoplasma pneumoniae* ✗

Causes atypical pneumonia, especially in school-aged children and young adults. It lacks a cell wall.

D. *Chlamydia pneumoniae* ✗

Can cause atypical community-acquired pneumonia but is less common than *S. pneumoniae*.

Key exam point

💡 Classic most common bacterial CAP → *S. pneumoniae*.

Remember

✓ Gram-positive diplococcus

✓ Encapsulated

✓ Lobar pneumonia

29. Which bacterium is a coliform commonly found in the gut, with a prominent capsule, lactose fermentation, non-motility, and oxidase negativity, and can cause pneumonia?

- A. *Salmonella typhi*
- B. *Klebsiella pneumoniae*
- C. *Shigella flexneri*
- D. *Vibrio cholerae*

Answer: B. *Klebsiella pneumoniae*

Klebsiella pneumoniae is a Gram-negative, encapsulated, lactose-fermenting, non-motile rod. It belongs to the Enterobacterales and can be found as part of the intestinal flora.

Its large polysaccharide capsule is an important virulence factor. *Klebsiella* can cause pneumonia, particularly in hospitalized or debilitated patients, and classically may produce thick, blood-tinged sputum.

Why Not Other Options?

A. *Salmonella typhi* ✗

Causes typhoid fever and is typically associated with enteric/systemic disease. It is motile and does not have the characteristic lactose-fermenting profile of *Klebsiella*.

C. *Shigella flexneri* ✗

Causes bacillary dysentery. It is non-motile but is a non-lactose fermenter.

D. *Vibrio cholerae* ✗

Causes cholera with profuse watery diarrhea. It is a curved/comma-shaped Gram-negative rod and is motile.

Key exam point

✦ Large capsule + lactose fermenter + non-motile → *Klebsiella pneumoniae*.

Remember

✓ *Klebsiella* → prominent capsule

✓ Lactose fermenter → pink colonies on MacConkey agar

✓ Non-motile + oxidase negative

30. Cell wall synthesis inhibitors are effective against the following organism EXCEPT:

A. *Streptococcus pneumoniae*

B. *Staphylococcus aureus*

C. *Mycoplasma pneumoniae*

D. *Treponema pallidum*

Answer: C. *Mycoplasma pneumoniae*

Mycoplasma pneumoniae does not have a cell wall. Therefore, antibiotics that work by inhibiting bacterial cell-wall synthesis, such as β -lactams, have no target and are ineffective against it.

This is an important distinguishing feature of Mycoplasma. It has a cell membrane containing sterols instead of a conventional peptidoglycan cell wall.

Why Not Other Options?

A. Streptococcus pneumoniae ✗

Has a peptidoglycan cell wall, so cell-wall-targeting drugs such as β -lactams can act on it when the strain is susceptible.

B. Staphylococcus aureus ✗

Has a thick Gram-positive peptidoglycan cell wall, making cell-wall synthesis an important antibiotic target.

D. Treponema pallidum ✗

Although it has a relatively unusual cell-envelope structure, it has peptidoglycan and is susceptible to penicillin, which interferes with cell-wall synthesis.

Key exam point

💡 No cell wall $\rightarrow$ β -lactams don't work $\rightarrow$ Mycoplasma pneumoniae.

Remember

✓ Mycoplasma $\rightarrow$ no cell wall

✓ Contains membrane sterols

✓ β -lactams $\rightarrow$ ineffective